

Obstruction certificates under incomplete observation

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October 3, 2026

Unrefereed author preprint; independent mathematical review pending.

Abstract

Under perfect measurement the viability kernel — the set of states from which some feedback keeps the system within its constraints — is characterized by tangency conditions. Under incomplete observation the sufficiency direction has a canonical answer in Veliov’s output-feedback regulation condition and the estimation-tube reduction. The complementary direction — certifying that no observation-based policy is viable — has received less systematic treatment.

This paper develops that instrument: a calculus of obstruction certificates, sound sufficient conditions for non-viability (equivalently, necessary conditions for viability), finitely checkable for declared polyhedral common-action and finite-system classes; the fibre criterion instead addresses static certification; analytic drift and timing tests have additional original-path premises. They do not exhaust the complement of the epistemic kernel, the kernel’s observation-based counterpart; in finite systems backward recursion is complete. The paper separates dynamic exit, epistemic action and timing obstructions, static certification limits, and failure of a declared policy class; these are distinct kinds of result, not one list of nonviability conditions. The central common-action obstruction shows that, at a declared information state under its held-action and local adverse-selection hypotheses, disjoint safe-control sets exclude a jointly safe observation-based policy, even when every compatible state is individually viable under full information. The others are a finite-time exit certificate conditional on an Isaacs-type drift bound and a compatible original-system adverse trajectory, an epistemic-emptiness construction in which a constant observation merges states whose only common admissible control is unsafe, a delayed-information obstruction with an explicit timing bound, and a certification limit: an exact observation-only certifier exists iff safe-set membership is constant on the observation fibres. The sixth, a certainty-equivalence trap, empties the kernel of the uncorrected controller under a biased observation even when the perfect-information kernel is nonempty. Two class-typed exact statements are established: over polytope-declared blind-window classes the timing certificate is exactly computable as a linear program, with the Farkas dual as the obstruction witness, with a separate

unbounded gap relative to a narrower whole-window hold class; and the residual dynamic gap is shown to decompose exactly, its post-observation recourse verdict not determined by the window sub-model alone.

The calculus is positioned against barrier certificates and estimation tubes, and its monitoring and institutional consequences — observation timing, coarseness, aggregation, and bias — are drawn.

Keywords: viability kernel; obstruction certificates; robust viability; epistemic viability; output feedback; sustainability governance; partial observation

Mathematics Subject Classification: 49J53; 93B03; 93C41; 91B76

1. Introduction

1.1 The problem

Viability theory characterizes the states of a constrained control system from which at least one control can keep every future state within a constraint set (Aubin, 1991). For sustainability problems the constraints are floors — stock levels, service thresholds, safety margins — and viability is the formal counterpart of maintaining a development path rather than merely optimizing it (Béné, Doyen, and Gabay, 2001; De Lara and Doyen, 2008; Doyen et al., 2012; Doyen and Gajardo, 2020). The theory under *perfect measurement* is complete in the relevant sense: the viability kernel exists as the largest viable subset, and Nagumo-type tangency conditions certify it.

Sustainability governance, however, operates under *incomplete observation*. Stocks are assessed at discrete review intervals. Indicators are coarser than the state. Some stocks are unobserved altogether. And the information that does arrive may arrive late. This paper addresses that side: for an incomplete observation structure, it develops conditional instruments for proving nonviability, separating information failures from dynamics-only exit and from static-verdict limits.

Floors in sustainability assessment are typically constraints on the productive base — the state of the system that yields the measured output — rather than on the output itself. Such a base can be read as natural capital, as a stock, or as a slowly regenerating flow of services; these are overlapping readings of the same asset rather than mutually exclusive ones, and what makes a use of it sustainable is whether the use falls on the yield or on the base itself. Across assets, whether the base behaves as a flow or as a stock is a continuum, set by its regeneration

timescale relative to the rate of use — a continuum that runs from a season, through a year, to geological time for a mineral deposit. If a base regenerates too slowly for the rate at which it is taken, drawdown is still liquidation.

A base can also erode while measured output is maintained, because use draws on the base rather than on its yield: the measured quantity can be held up while the base declines, and partial observation may not detect the erosion until a floor is breached. The signal that would reveal the erosion is accordingly often the one that is coarse, postponed, or absent. Incomplete observation is therefore structural to the certification problem, not merely a practical limitation; the sufficiency literature and the obstruction calculus of this paper address the same underlying indeterminacy rather than competing ones.

The question splits naturally. The sufficiency direction has a canonical literature: Veliov (1993) gives conditions for the existence of an output-feedback regulation map under incomplete and inexact measurement, reducing under perfect measurement to the classical viability condition (Haddad, 1981); the estimation-tube programme absorbs imperfect measurement into set-valued dynamics on an estimation space of sets of compatible states (Kurzanski and Vályi, 1997; Cardaliaguet, Quincampoix, and Saint-Pierre, 2007), a reduction that is exact only under its own matching policy and admissibility hypotheses; and barrier certificates certify safety (Prajna and Jadbabaie, 2004; Prajna, Jadbabaie, and Pappas, 2007), with converse and hybrid necessary-and-sufficient characterizations (Prajna and Rantzer, 2005; Maghenem and Sanfelice, 2019). What that literature does not supply is the converse direction: a calculus of obstruction certificates, each a sufficient condition for nonviability whose negation is a necessary condition for observation-based viability. An obstruction certificate is a checkable witness that no observation-based policy exists — finite in the finite-state and polyhedral cases — and the argument it makes is that a prescribed class of observation-based policies fails not because a particular policy is bad, but because the information structure leaves no room for any policy. In the polyhedral common-action and certification forms (Theorem 2 and Proposition 7) it is a finite, checkable test; the drift certificates of Theorem 5 and Theorem 3 are not finite objects, one exhibiting an enforcement selection and the other bounding the timing uniformly over every policy. Veliov’s condition tells us when output feedback can work; the obstruction calculus tells us when it cannot, and for the borderline cases it identifies which quantitative feature of the observation design — timing, coarseness, bias, aggregation — is responsible.

1.2 Contributions

The following certificates, classification criteria, exact algorithms and policy-class cautions play different roles:

1. **The instantaneous common-action obstruction** (*finitely checkable*) (Theorem 2, Example 1). If the safe-control sets of the states compatible with the current information set intersect emptily, then no observation-based policy is viable, even though every compatible state is individually viable under full information. The failure is purely informational: no stochasticity, no estimation error.
2. **The delayed-information obstruction** (*closed-form conditional*) (Theorem 3). If the information needed to distinguish safe from unsafe compatible states arrives only at time T_{obs} , and the disturbance can force constraint violation earlier than T_{obs} , then no observation-based policy is viable. The timing bound $\inf q/\varepsilon < T_{\text{obs}}$ is stated explicitly: information may be accurate but arrive too late.
3. **The finite-time exit certificate** (*closed-form conditional*) (Theorem 5). Under a Dini-drift condition of Isaacs type — the disturbance chooses after the control — the disturbance can force exit from the safe set within an explicit time bound, for *every* admissible control of any information structure. This is the base obstruction: when it applies, the viability question is closed without any observation-theoretic argument.
4. **Epistemic emptiness by admissibility** (*minimal construction*) (Proposition 5, a minimal construction). A constant observation merges states whose *admissible* controls differ: the state-dependent control set forces a single common action, and that action is unsafe at the boundary, so the unresolved initial observation fibre lies outside the epistemic kernel, even though every physical state is viable under full information. The mechanism is admissibility — the common-action obstruction of Theorem 2 at minimal size — and Example 1 gives the hidden-mode instance.
5. **The fibre certification criterion** (*finitely checkable*) (Proposition 7, Corollary 1). An exact observation-only safety certifier — a function of the observation that returns “safe” exactly on the safe set — exists if and only if safe-set membership is constant on every observation fibre. The certainly-safe set is the sound relaxation: the largest set of observations that can be labeled safe without further information.
6. **The certainty-equivalence trap** (*policy-class restriction*) (Remark 5). Even an *injective* observation empties the kernel of a system whose perfect-information kernel is nonempty for the canonical certainty-equivalence controller — the perfect-information zeroing law applied to the uncorrected observation. The trap is a policy-specific failure: a controller that uses the observation map’s structure (the bias correction) restores viability.
7. **Computable instantiation and the exact residual** (*finitely checkable; exact residual*) (Theorem 4, Example 2, Propositions 9 and 10). Over polytope-declared blind-window classes the timing certificate is exactly computable as a linear program — polynomial size, exact in rational arithmetic, Farkas-certified on failure — sound beyond the declaration, with the implementable-class gap shown unbounded by an explicit instance on which the relaxation extinguishes the certificate. The residual gap of the completeness question decomposes exactly into the window phase, decided completely by the certificates, and the post-observation recourse phase, decided by no window-measurable certificate pair.

The calculus is organized by a single selector principle (Proposition 1, Section 2.4): observation-based viability is the existence of one response — admissible, safe, and recursively viable — for every compatible state, and each certificate proves that a specific response correspondence is empty; the obstructions are nested in a ladder (Proposition 3). Five refinements complete the picture: backward belief recursion is sound and complete on any finite horizon in finite systems (Theorem 1); a uniform-margin condition connects the instantaneous and tube obstructions (Proposition 2); the timing obstruction has a sharp threshold form (Remark 2); a Helly-type sparse witness bounds the common-action obstruction by $m+1$ compatible states (Proposition 4); and a comparison-function form bounds first boundary contact under its path assumptions (Remark 4). The epistemic kernel is monotone in action sets, disturbances, information, and policy class (Proposition 8).

Throughout, “obstruction” means a *certified* failure. Theorem 5 and Theorem 3 each exhibit the violating constraint, an admissible disturbance, and the quantitative bound that make failure inevitable, for every policy in the prescribed class; Proposition 5 and Theorem 2 exhibit the incompatible admissible controls — the merged observation whose incompatible safe controls certify emptiness — with no quantitative bound claimed there; and Proposition 7 is the converse-side characterization of the certification limit itself. The calculus is positioned against the barrier-certificate literature (Section 6.2) and the estimation-tube programme (Section 6.3), and its consequences for the design of monitoring and indicators in sustainability governance are drawn in Section 6.4.

1.3 Related work

The viability-theory background is Aubin (1991), Aubin, Bayen, and Saint-Pierre (2011), and the kernel approximation theory of Saint-Pierre (1994); robust viability is Aubin and Frankowska (1990) and Frankowska (1989). Under incomplete measurement, the sufficiency theorem is Veliov (1993), viability with a priori unknown but observable parameters is Quincampoix and Veliov (1994), and the estimation-set reduction is Cardaliaguet, Quincampoix, and Saint-Pierre (2007). Doyen (2000) studies output feedback for uncertain nonlinear systems under state and control constraints and obtains necessary and sufficient conditions for a property close in spirit to ours: a set K is a *guaranteed viability domain* when there exists a Lipschitz selection $u(\cdot)$ of the control constraint map such that K is invariant under the differential inclusion $\dot{x} \in \{f(x, u(h(x, w))), w) : w \in W(x)\}$, the disturbance acting on both the dynamics and the output; his characterization is by non-emptiness, for some $\lambda > 0$, of the Lipschitz kernel $L_\lambda(R_K^\lambda)(y)$ at every output y , and a viable feedback is synthesized by Steiner selection. Doyen’s result is the nearest neighbour to the question posed here, and it differs from it on three axes, each visible in his own statement. First, the *policy class*: the definition quantifies over Lipschitz selections, and the closed loop is $u(h(x, w))$, a memoryless function of the current output — so that a policy which is non-Lipschitz, or which uses the history of observations rather than the current output alone, lies outside the range of his necessity. The

obstruction certificates are asserted for observation-based policies generally and are neither implied by nor subsumed under his condition. Second, the *property*: his is exact invariance of a closed set, one particular viability property among those addressed here. Third, the *direction*: the characterization is an equivalence whose constructive content is a feedback-building corollary, while the systematic obstruction side — certifying that no observation-based policy is viable — is not developed there, and Doyen records the reason: “to our knowledge, no general viability result is available in this differential game context with imperfect and/or partial information.” It is that undeveloped side, and not the synthesis, that the obstruction calculus is meant to serve. On the failure side, Aubin (2001) characterizes the complement of the kernel — Poincaré’s “shadow” — together with the capture basins, and Aubin and Catté (2002) supply the bilateral-fixed-point and discriminating-kernel calculus; the certificates of Section 3 are one-sided, observation-constrained analogues of that programme. In verification, barrier certificates originate with Prajna and Jadbabaie (2004), with worst-case and converse results in Prajna, Jadbabaie, and Pappas (2007) and Prajna and Rantzer (2005) and hybrid characterizations in Maghenem and Sanfelice (2019). The dominant computational tradition in safety verification reaches the same question from the opposite side. Hamilton–Jacobi reachability formulates safety as a two-player differential game and recovers the backward reachable tube as the zero sublevel set of the viscosity solution of a time-dependent Hamilton–Jacobi–Isaacs equation (Mitchell, Bayen, and Tomlin, 2005; Margellos and Lygeros, 2011; for a survey, Bansal, Chen, Herbert, and Tomlin, 2017), the analytic foundation being the viscosity-solution theory of Crandall, Ishii, and Lions (1992). It does not reach the setting considered here, for a reason that is structural rather than technical. The value function is posed on the physical state $x \in \mathbb{R}^n$, and it is the state space itself that the grid — or the collocation points of a neural solver — discretizes. Under incomplete observation the sufficient statistic is not a state but a belief, a probability measure over states, and the value function would accordingly have to live on an infinite-dimensional space of measures; level-set and physics-informed solvers are discretization schemes for finite-dimensional domains and do not carry over. The obstruction calculus sidesteps the difficulty by not solving a differential equation at all: the certificates of Section 3 are finite algebraic objects, checkable without discretizing any continuum. Two further contrasts are worth recording. Even with perfect observation the Hamilton–Jacobi certificate is only as exact as its discretization: grid-based solvers scale exponentially in the state dimension and are in practice confined to five or six dimensions, and the learning-based approaches that lift that bound (Bansal and Tomlin, 2021) purchase scalability with approximation — their grid-referenced benchmark, a three-dimensional pursuit–evasion game, reports a 0.43% backward-reachable-tube volume error against the grid solution, and at six dimensions and above, where such methods are needed, no grid reference exists at all: the six-dimensional instance is validated only by projection back to relative coordinates. And in *direction* the tradition is, like viability theory proper, a sufficiency machine: it computes the states from which some control against

all disturbance preserves safety, which is precisely the direction Veliov’s condition and the estimation-tube reduction already serve. The obstruction certificates supply the complementary direction, and do so in the observation-constrained regime in which neither tradition returns an answer. The sustainability application domain is Béné, Doyen, and Gabay (2001), De Lara and Doyen (2008), Doyen et al. (2012), and Doyen and Gajardo (2020). To our knowledge, the obstruction certificates of Theorem 2, Theorem 3, and Proposition 7 — in particular the certification criterion and the timing bound — have not been stated in this form; the elementary facts on which they rest (quantifier commutation and Dini comparison) are classical. Relative to Doyen (2000) the residual contribution is correspondingly exact, and is the content of the three distinctions above: a certificate that is checkable, that holds against *every* observation-based policy in the declared class under its realization and blind-record hypotheses — not merely against Lipschitz memoryless selections of the current output — and that applies to viability properties beyond the exact maintenance of a closed domain. It is in that sense that the obstruction calculus does not compete with his characterization but occupies the side of it that he leaves open.

1.4 Organization

Section 2 fixes the framework: the control system, the viability kernel hierarchy, observation structures, information states, the safe-control correspondence, and the selector principle (Proposition 1). Section 3 develops the obstruction calculus (Theorem 1–5, Propositions 2–5, Remarks 2–4, and Example 1): finite-horizon completeness, the instantaneous common-action obstruction with its ladder and sparse witness, the delayed-information obstruction with its threshold form, and — as supporting certificates — the finite-time exit certificate and the admissibility obstruction, together with a recourse certificate for the post-observation mode (Proposition 6, Example 3). Section 4 develops the certification limits (Proposition 7, Corollary 1, Remark 5). Section 5 reviews the sufficiency landscape, cited rather than re-derived. Section 6 discusses the position of the calculus relative to barrier certificates and estimation tubes, states the monotonicity of the epistemic kernel (Proposition 8), and draws the governance consequences. Sections 7–10 develop the new material: complete characterizations and the residual gap (Section 7), the selector principle with its dual witnesses and the monitoring-adequacy converse (Section 8), a symbolic case study with a coverage audit (Section 9), and the probabilistic belief-state development (Section 10). Section 11 concludes. Brief sketches and short proofs of the main-text statements are given in place; the extended discussions and worked instances of the Sections 3, 4, and 10 certificates, the complete proofs of the Section 3 certificates, the expanded sufficiency review, and the auxiliary constructions are collected in the Supplementary Material; the results of Sections 7 and 9 are proved in the main text. The running instances are fixed here for orientation: the two-floor conflict (Supplementary S1), on which the worked instances of the Sections 3, 4, and 10 certificates are built, and the symbolic delayed-hidden-regime case study (Section 9).

2. Framework

2.1 The control system and the kernel hierarchy

This section fixes the mathematical objects used throughout the paper: the control system, the disturbance structure, the constraint set, and the four kernels (perfect-information, robust, epistemic, and institutionally restricted) that organize the analysis.

Fix a state space $X \subseteq \mathbb{R}^n$, a control set U , a disturbance set D , and dynamics

$$\dot{x}(t) = f(x(t), u(t), d(t)), \quad u(t) \in U(x(t)), \quad d(t) \in D(x(t)),$$

with f continuous, U and D set-valued with closed graph and nonempty compact values, and admissible controls and disturbances taken to be measurable selections — the standing solution-existence conditions: every selection pair admits a Carathéodory solution, and the convexified relaxed inclusion of Section 3.5 satisfies the Marchaud conditions of the viability literature (Aubin, 1991). Let $\mathcal{V} \subseteq X$ be a closed constraint set. A *causal policy* π is a map from the observation record to controls, respecting the specified information structure. The standard objects are:

- $\text{Viab}(\mathcal{V}; U, \pi_{\text{perf}})$: the **viability kernel** under full-information (state-feedback) policies — the largest closed subset of $\mathcal{V}$ from which there exists a state-feedback control keeping the trajectory in $\mathcal{V}$ for all time — no disturbance, or a known disturbance trajectory (Aubin, 1991).
- $\text{RViab}(\mathcal{V})$: the **robust kernel**, the largest subset of $\mathcal{V}$ from which there exists a state-feedback policy keeping every trajectory — for *every* admissible disturbance realization — in $\mathcal{V}$ (Aubin and Frankowska, 1990).
- $\text{ERViab}_{\mathcal{I}}(\mathcal{V})$: the **robust epistemic kernel** — the object of this paper (Definition 1). Its elements are *states of information*, not physical states (Section 2.3). The non-robust counterpart $\text{EViab}_{\mathcal{I}}(\mathcal{V})$ is the non-robust contrast class (Section 2.3).

Projected to physical state space — with $\text{proj}_{\exists}(\mathfrak{B}) = \bigcup_{B \in \mathfrak{B}} B$ the *existential* projection of a collection $\mathfrak{B}$ of information states, applied to the first two terms, whose elements are information states — the informational hierarchy reads

$$\begin{aligned} \text{proj}_{\exists}(\text{IRViab}_{\mathcal{I}}(\mathcal{V})) &\subseteq \text{proj}_{\exists}(\text{ERViab}_{\mathcal{I}}(\mathcal{V})) \\ &\subseteq \text{RViab}^{\text{hist}}(\mathcal{V}), \end{aligned}$$

where $\text{IRViab}_{\mathcal{I}}(\mathcal{V})$ is the institutionally restricted kernel of Section 6.4 and $\text{RViab}^{\text{hist}}$ permits nonanticipative full-information histories. The relation to the state-feedback RViab and nondisturbed Viab requires matching admissibility, solution and policy classes; no equality is inferred here. The physical implication of epistemic viability is the inclusion at the level of beliefs: $B_0 \in \text{ERViab}_{\mathcal{I}}(\mathcal{V}) \Rightarrow B_0 \subseteq \text{RViab}(\mathcal{V})$ only when observation-based nonanticipative strategies admit a full-information state-feedback realization with the same timing and disturbance quantifiers. Without that comparison theorem, the inclusion is into the full-history robust viability set, which need not have been identified with the state-feedback kernel. Each inclusion can be strict, with a distinct cause: robust contraction arises from disturbances; epistemic contraction from indistinguishability; institutional contraction from restricted authority, enforcement, and allocation. The present paper characterizes the *epistemic* contraction: the mechanisms by which indistinguishability empties or shrinks the kernel.

2.2 The safe-control correspondence

For a constraint set $\mathcal{V}$ described by finitely many C^1 constraint functions, $\mathcal{V} = \{x : q_j(x) \geq 0, j = 1..m\}$, define the **safe-**

control correspondence at $x \in \mathcal{V}$:

$$\mathcal{R}_{\mathcal{V}}(x) = \{u \in U(x) : \forall d \in D(x), \forall j \text{ with } q_j(x) = 0, \\ \nabla q_j(x) \cdot f(x, u, d) \geq 0\},$$

the set of controls whose drift is inward (subtangent) on every active constraint face, uniformly over the disturbance. For x in the interior of $\mathcal{V}$ (no active constraints), $\mathcal{R}_{\mathcal{V}}(x) = U(x)$. Two levels of the tangency condition are distinguished throughout. (i) *Local reading*. For the finite C^1 representation $\mathcal{V} = \{q_j \geq 0\}$, every contingent tangent velocity satisfies the active-gradient inequalities. The reverse inclusion, and hence a sufficient Nagumo reading of $\mathcal{R}_{\mathcal{V}}$, requires a representation regularity condition, for example the Mangasarian–Fromovitz constraint qualification at the relevant boundary points, together with the usual solution and measurable-selection assumptions. Without such a qualification the gradient inequalities are necessary but not sufficient for tangency: $q(x) = x^3$, $\mathcal{V} = [0, \infty)$, and velocity -1 at $x = 0$ satisfy $\nabla q(0) \cdot (-1) = 0$ yet exit immediately. By contrast, a *strictly negative* derivative at an active constraint along an admissible original-system path gives a local obstruction without a constraint qualification. These two directions must not be conflated. (ii) *Kernel reading*. When $\mathcal{V}$ is not invariant, viability of a point $x_0 \in \mathcal{V}$ requires instead a selection $u(x) \in \mathcal{R}_G(x)$ along the trajectory, where $G = \text{RViab}(\mathcal{V})$ and $\mathcal{R}_G$ is the same correspondence computed on the kernel: points of $\mathcal{V} \setminus G$ may admit controls that are instantaneously safe while every trajectory must exit later. The obstruction theorems below are *local certificates against* $\mathcal{R}_{\mathcal{V}}$: exhibiting a state at which even the weaker pointwise condition fails certifies exclusion from the kernel, and this is the reading in which the certificates are used. The correspondence is the bridge between the pointwise (tangency) and the global (kernel) description, and it is the object the observation structure acts upon: an observation-based policy can select controls only as a function of the observation record.

2.3 Observation structures, information states, and epistemic kernels

Conventions. The plant evolves in continuous time; sampling and hold are hypotheses only of those statements that explicitly declare a review and a held first command. A continuous record may be available between reviews, and a time-varying control is permitted when the declared policy class includes it. The observation record is generated by O ; error, hidden-parameter and delay variants need their own information model. A policy may use the full nonanticipative observation record. Representing it solely by the set-valued belief B_t is justified when the exact belief is a sufficient statistic and a measurable policy selector exists for the stated strategy class, not by the definition of a record alone. In a blind window, observation-equivalent branches share the same control, but this does not require that control to be constant unless a hold is declared. Disturbance information order is statement-specific: the common-action and delayed witnesses respond to a candidate policy; the oracle-recourse witness uses one exogenous scenario admissible against all policies. An action outside $U(x)$ on a compatible state is failure; strict floor violation means $q < 0$.

An **observation structure** $\mathcal{I} = (Y, O)$ is a measurable space Y and an observation map $O : X \rightarrow Y$. The policy observes $y(t) = O(x(t))$. At time t , the **information set** (belief) of a regulator who knows the dynamics, the disturbance class, its own applied controls $(u(s))_{s < t}$, and the record $(y(s))_{s \leq t}$ is

$$B_t = \{\xi(t) : \xi(0) \in B_0, \\ \dot{\xi}(s) = f(\xi(s), u(s), d(s)), d(s) \in D(\xi(s)), \\ O(\xi(s)) = y(s) \text{ for all observed } s \leq t\},$$

where the applied control history $u(\cdot)$ in the dynamics is the regulator’s own — common to every candidate trajectory — and $\xi(\cdot)$ ranges over candidate trajectories: the set of current states compatible with the record and the applied controls. Under the stated set-membership semantics (Kurzhanski and Vályi, 1997), B_t contains every state consistent with the observations and applied controls. A one-step exact update equals reachable states intersected with the reading cell only if admissible original paths can be restricted and glued, disturbances are rectangular and memoryless across the step, and the surviving paths can be extended as required. Otherwise that formula is an outer approximation, not the exact posterior. Closed observation cells and compact reachable sets are separate prerequisites for compact posteriors; measurability of a viable belief-policy selector is another separate prerequisite. (With observation error, the equality becomes membership $y(s) \in H(\xi(s), v(s))$; with hidden parameters the belief lives in $X \times \Theta$; with delays it lives in a history space — the extensions are noted as open in Section 6.5.) A policy is **observation-based** when it is a measurable nonanticipative function of the available record. A **belief-based** policy factors through B_t , and is a special case. Replacing arbitrary record policies by belief policies requires exact sufficient-statistic semantics and an implementable measurable selector; it is not guaranteed merely by delay-free measurement. We write $U^B(B) = \bigcap_{x \in B} U(x)$ for the controls available at every compatible state — the only controls an observation-based policy may select without risking inadmissibility.

Definition 1 (robust epistemic kernel). $B_0 \in \text{ERViab}_{\mathcal{I}}(\mathcal{V})$ if there exists a nonanticipative observation-based policy π such that, for every compatible initial state $x_0 \in B_0$ and every admissible disturbance realization $d(\cdot)$, the trajectory of $\dot{x} = f(x, u(t), d(t))$ generated from x_0 by π ’s actions under $d(\cdot)$ — the record being the one that $(\pi, d(\cdot))$ generates — remains in $\mathcal{V}$ for all time. (This is the standard exists-strategy-for-all-realizations form. The quantifier order is essential: the record is generated by the realization, so conditioning the admissible realizations on the record would be circular.) The common ambient domain for this kernel, for every sensor, is $\mathfrak{B} = \{B \subseteq X : B \neq \emptyset\}$; compactness is assumed separately where a theorem needs it. A general B_0 is a *prior* before the initial observation; the initial reading $y_0 = O(x_0)$ is available before the first action. The resulting current information cell is $B_{0,y} = B_0 \cap O^{-1}(y)$, and subsequent beliefs similarly lie in one current observation cell. Only such a post-observation cell shares one action $u \in U^B(B)$. For a prior spanning readings, the policy may choose separate actions for different y , so the common-action theorems are applied to its nonempty cells, not to the unsplit prior. Full fibres are the initial cells when $B_0 = X$; histories can yield strict subsets. Observation measurability alone does not imply compact fibres; the domain does not silently impose this property. The existential (non-robust) counterpart $\text{EViab}_{\mathcal{I}}(\mathcal{V})$ is the non-robust contrast class.

We use ERViab throughout, since the disturbance classes of sustainability problems are adverse by construction. The theorems are stated for this robust notion; they do not transfer a fortiori to the non-robust contrast class $\text{EViab}_{\mathcal{I}}(\mathcal{V})$, in which there exist an observation-based policy and *some* admissible disturbance realization under which every compatible trajectory remains in $\mathcal{V}$ for all time. Since the robust kernel is contained in the non-robust one, an observation-based policy may exist there that no robust policy survives.

Under exact rectangular belief dynamics and a strategy class admitting nonanticipative concatenation, the epistemic kernel is a recursively viable family of information states. Without those hypotheses, this is an organizational description rather than an exact greatest-fixed-point theorem. In every class, common-action compatibility is a necessary one-step condition,

but not by itself a complete characterization — and the obstruction calculus certifies, mechanism by mechanism, some information states outside it under their stated hypotheses.

2.4 Notation

We collect the principal symbols used throughout the paper and fix the two structural symbols: the **observation structure** is always $\mathcal{I}$ (calligraphic I) and the **institution** of Section 6.4 is always $\mathcal{J}$ (calligraphic J); no other letter serves either role. The state space is $X \subseteq \mathbb{R}^n$; the control set is U (set-valued $U(x)$ at state x); the disturbance set is D (set-valued $D(x)$). The dynamics are $\dot{x} = f(x, u, d)$. The closed constraint set is $\mathcal{V} \subseteq X$, described by finitely many C^1 constraint functions q_j ($j = 1, \dots, m$). The safe-control correspondence at $x \in \mathcal{V}$ is $\mathcal{R}_\mathcal{V}(x)$. The observation structure is $\mathcal{I} = (Y, O)$ with observation map $O : X \rightarrow Y$; the information set (belief) at time t is B_t ; the initial belief is B_0 — a prior in the common ambient domain $\mathfrak{B}$ which is split by the initial observation into nonempty post-observation cells $B_{0,y} = B_0 \cap O^{-1}(y)$. The common admissible action set at belief B is $U^B(B) = \bigcap_{x \in B} U(x)$ — the only controls an observation-based policy may select without risking inadmissibility, under the convention, used from Proposition 5 on, that an action outside $U(x)$ at a compatible state is itself failure; the common safe-action set is $\mathcal{R}_\mathcal{V}^B(B) = \bigcap_{x \in B} \mathcal{R}_\mathcal{V}(x)$. The tube-safe action set at review length Δ is the set $\mathcal{A}_{\text{tube}}(B, \Delta)$ of commands $a \in U^B(B)$ for which *every* compatible admissible original-system held-action trajectory exists, remains in $\mathcal{V}$, and obeys $a \in U(x(t))$ throughout $[0, \Delta]$. The last condition matters when admissible controls depend on state; initial common admissibility does not imply trajectory-wise admissibility. Statements about its emptiness apply only to a policy class required to hold the first action for that Δ . The first informative observation time is T_{obs} : the first time the observation record discriminates between compatible branches — before it, the branches of Theorem 3’s (H3.1) produce identical records. The adverse-selection correspondences of the exit certificates are $D_\varepsilon(x, u) = \{d \in D(x) : D^+q(x; f(x, u, d)) \leq -\varepsilon\}$ (Theorem 5) and $D_\eta(x) = \{d \in D(x) : \nabla q_j(x) \cdot f(x, a, d) \leq -\eta/2\}$ (Theorem 2’s proof, local). The upper right Dini derivative of q in direction v is $D^+q(x; v) = \limsup_{h \downarrow 0} [q(x + hv) - q(x)]/h$; for q of class C^1 — the class assumed in the exit theorems — $D^+q(x; v) = \nabla q(x) \cdot v$.

The kernels of the hierarchy are $\text{Viab}(\mathcal{V}; U, \pi_{\text{perf}})$ (perfect information), $\text{RViab}(\mathcal{V})$ (robust), and $\text{ERViab}_\mathcal{I}(\mathcal{V})$ (the robust epistemic kernel — Definition 1, the object of this paper), with $\text{proj}_\mathfrak{B}(\mathfrak{B}) = \bigcup_{B \in \mathfrak{B}} B$ the existential projection of a collection $\mathfrak{B}$ of information states. When the observation O is injective, the record determines the state, but equality with a state-feedback robust kernel requires an additional policy-class reduction; under such a reduction the corresponding kernels agree, the informational certificates of Sections 3.2–3.3 and 3.6 reduce to the no-merge case, and only the dynamic obstruction of Section 3.5 can certify exclusion — the definitions degenerate to classical robust viability exactly when observation is perfect. The non-robust counterpart $\text{EViab}_\mathcal{I}(\mathcal{V})$ is the non-robust contrast class (Section 2.3). The institutionally restricted kernel is $\text{IRViab}_\mathcal{J}(\mathcal{V})$ (Section 6.4). The policy classes are π_{perf} (state feedback), Π_{output} , Π_{state} , and Π_{CE} — the certainty-equivalence class defined in Remark 5: fixed perfect-information regulation laws applied uncorrected to the observation. The policy-specific safety set of Section 4.1 is $\text{Safe}_\mathcal{I}^\Pi$; the certainly-safe set of Corollary 1 is $\mathcal{Y}_{\text{safe}}$.

A few letters carry local meanings, as declared here: the strip width a of Theorem 5’s $\mathcal{S}_a$ is not the held action $a = \pi(B)$ of Theorem 2’s proof, nor the pathway vector $a \geq 0$ of Section 5(d); λ is the Farkas multiplier of Theorem 2’s checkability certificate and the observer decay rate of Section 5(c); and α is the

comparison function of Remark 4 and a multiplier of Section 5(d)’s substitution alternative. The letter K denotes the safe set of Section 4’s certification problems, while in Section 5(c) it denotes the kernel itself, with the buffered set K_ζ and eroded kernels K^{-c_ζ} of Section 5(c). The observation-error map of Section 2.3 is $H(\xi, v)$, whereas the harvest vector of the Supplementary Material is the lowercase h ; and the finite-horizon recursion set $\mathcal{W}_N$ of Section 3.1 is calligraphic, distinct from the Lyapunov function $W = S_1 + S_2$ of the Supplementary Material.

Shared symbols across the companion papers. The companion papers (worked systems, computational certification, probabilistic sufficiency, exact belief computation, minimax dual certificates, applied regime viability, and the applied forecast ladder) deliberately share one formal vocabulary; the letters below carry paper-local meanings here that differ there:

Here	In the companions
K : the safe set of Section 4’s certification problems; the kernel of Section 5(c)	window depth and contact set $\mathcal{K}(B)$ (minimax duals); the full-information kernel K_{full}^T (certification); stock carrying capacities (forecast ladder; minimax benchmark)
λ : the Farkas multiplier of Theorem 3’s certificate; the observer decay rate of Section 5(c)	certificate weight vectors (certification; minimax duals)
Γ_h : the recourse certificate of Section 5(d)	the certificate margin $\Gamma_\mathcal{I}$ (certification); the alpha-vector witness sets Γ_k^Π (sufficiency)
τ : delay and observation times	the worst-case exit time τ^* (worked systems)
σ : the recourse construction’s trajectory parameter	the kernel time variable (certification; minimax duals); the floored exit count σ^* (worked systems); a noise scale (forecast ladder)
μ : a Farkas multiplier	an adversarial probability measure (certification; minimax duals)

The continuous response correspondences and separately typed discrete predecessor provide one organizing skeleton, not an unrestricted completeness theorem. An observation-based regulator selects a single response for an entire information set — a held action, a feedback segment, or a safe/unsafe verdict — and observation-based viability is the existence of a response that is simultaneously admissible, safe, and recursively viable for every compatible state and disturbance history. Each dynamic nonviability certificate of Section 3 is a sufficient condition under its declared policy and original-path hypotheses for failure of a response correspondence. Section 4 instead concerns static verdicts and a specified uncorrected feedback law. The dynamic correspondences include: $U^B(B)$ (admissibility), $\mathcal{R}_\mathcal{V}^B(B)$ (instantaneous safety), $\mathcal{A}_{\text{tube}}(B, \Delta)$ (review-interval safety), or the recursive predecessor of Section 3.1. The mechanisms differ in which correspondence fails, and the obstructions are nested (Proposition 3). The organizing claim of the calculus is the following, stated once and used implicitly by every certificate of Sections 3 and 4.

Proposition 1 (selector principle). If $B \in \text{ERViab}_\mathcal{I}(\mathcal{V})$ is a *post-observation information cell* and π is a policy witnessing viability on B , then at every review time the action $a = \pi(B)$ selected by π is a response that is (i) *admissible*, $a \in U^B(B)$; (ii) *safe*, keeping every compatible trajectory in $\mathcal{V}$ until the next observation — $\text{Reach}_{[0, \Delta]}(B, a) \subseteq \mathcal{V}$ at review length Δ ;

and (iii) *recursively viable*, in that every post-observation belief reached at the next review is again in $\text{ERViab}_{\mathcal{I}}(\mathcal{V})$ under the continuation of π . A belief is therefore viable only if each of the response correspondences of this section — $U^B(B)$, $\mathcal{R}_{\mathcal{V}}^B(B)$, $\mathcal{A}_{\text{tube}}(B, \Delta)$, and the recursive predecessor of Section 3.1 — is nonempty along every reachable branch; hence a certificate that any one of them is empty at a reachable belief certifies nonviability. In finite systems the converse is exact (Theorem 1); in continuous time no general finite backward-recursion converse is claimed (Section 6.5).

3. The Obstruction Calculus

3.1 Finite-horizon completeness (finite systems)

In the finite setting the calculus is complete: backward recursion over beliefs — the set-membership analogue of the belief-state recursion of partially observable Markov processes (Smallwood and Sondik, 1973) — is sound and complete on any finite horizon, and nonviability is certified by a finite obstruction tree. The certificates of Sections 3.2–3.3 and 3.5–3.6 are sufficient conditions, not a complete characterization of the epistemic kernel.

Fix finite state, action, disturbance, and observation spaces X, A, D, Y , a transition map $x^+ = F(x, a, d)$, an observation map $y = O(x, a, d, x^+)$, a safe set $\mathcal{V} \subseteq X$, and admissible-action sets $U(x) \subseteq A$. For a post-observation belief $B \subseteq X$ contained in one current observation cell, an action a , and a successor observation y , the post-belief is

$$\text{Post}(B, a, y) = \{x^+ \in X : \exists x \in B, d \in D(x), \\ x^+ = F(x, a, d), y = O(x, a, d, x^+)\},$$

and y is possible under (B, a) if $\text{Post}(B, a, y) \neq \emptyset$. With $U^B(B) = \bigcap_{x \in B} U(x)$, define the one-step predecessor of a collection $\mathcal{C}$ of beliefs by

$$\text{Pre}(\mathcal{C}) = \{B \subseteq \mathcal{V} : \exists a \in U^B(B) \text{ such that} \\ \text{Post}(B, a, y) \in \mathcal{C} \text{ for every possible } y\},$$

and the finite-horizon kernels by $\mathcal{W}_0 = \{B : B \subseteq \mathcal{V}\}$, $\mathcal{W}_{k+1} = \text{Pre}(\mathcal{W}_k)$. The N -step common safe-action set at a belief B is $\mathcal{A}_N(B) = \{a \in U^B(B) : \text{Post}(B, a, y) \in \mathcal{W}_{N-1} \text{ for every possible } y\}$, so that $B \in \mathcal{W}_N \iff \mathcal{A}_N(B) \neq \emptyset$. The recursion is exact but of worst-case complexity exponential in $|X|$ — the belief space contains up to $2^{|X|}$ beliefs — so finite-checkability is per belief, not a polynomial-time global procedure.

Theorem 1 (finite-horizon soundness and completeness). An initial *post-observation cell* B_0 admits an observation-based policy keeping every compatible trajectory in $\mathcal{V}$ for N steps if and only if $B_0 \in \mathcal{W}_N$. For a prior $P \in \mathfrak{B}$ before the initial reading, the verdict is $P \in \text{ERViab}_{\mathcal{I}, N}$ if and only if every nonempty cell $P \cap O^{-1}(y)$ belongs to $\mathcal{W}_N$.

Proof sketch. Both directions are inductions on the horizon. Necessity: a policy safe for N steps picks $a \in U^B(B_0)$, and every possible successor belief is safe for $N - 1$ steps, so $B_0 \in \text{Pre}(\mathcal{W}_{N-1})$. Sufficiency: choose the witness action at each step, and every reachable state remains in $\mathcal{V}$ for N steps. If $B_0 \notin \mathcal{W}_N$, the failure is certified by a finite obstruction tree alternating regulator and nature turns — the complete construction, with the one-step abstraction of Example 1 as its smallest tree, is given in the Supplementary Material (S1; the tree is drawn in Figure S2).

3.2 The instantaneous common-action obstruction

The instantaneous obstruction is the static core of the calculus: it certifies nonviability at a single information state, without a timing or horizon argument.

Theorem 2 (common-action obstruction). Let $B \subseteq \mathcal{V}$ be an information set of the specified observation structure (the safe-control correspondence $\mathcal{R}_{\mathcal{V}}$ is defined on $\mathcal{V}$, so the belief is taken inside it). Suppose the following hypotheses hold.

(H2.1) The selected action is held until the next observation (sampled review: no informative observation arrives before the next action must be selected, and the policy holds the selected action over the review interval — the institutional reading; without a positive dwell time the instantaneous statement below must be replaced by the tube-safety statement below).

(H2.2) Either the **common admissible action set** is empty,

$$U^B(B) = \bigcap_{x \in B} U(x) = \emptyset, \quad (\text{admissibility obstruction})$$

or the **common safe-action set** is empty while common admissibility holds,

$$\mathcal{R}_{\mathcal{V}}^B(B) := \bigcap_{x \in B} \mathcal{R}_{\mathcal{V}}(x) = \emptyset, \quad (1)$$

with $U^B(B) \neq \emptyset$ (safety obstruction)

(H2.3) **Original-system adverse trajectory (safety case).** For each $a \in U^B(B)$ in the safety case there exist $x_a \in B \cap \partial \mathcal{V}$, an active constraint j_a , numbers $c_a, \tau_a > 0$, and an admissible original-system Carathéodory pair $(x_a(\cdot), d_a(\cdot))$ from x_a under the held action a , compatible with the same pre-review observation record, such that $d_a(t) \in D(x_a(t))$ almost everywhere and

$$q_{j_a}(x_a(t)) \leq -c_a t < 0 \quad (0 < t < \tau_a).$$

No lower semicontinuity inheritance for a thresholded disturbance correspondence, measurable feedback $d(x)$, or convexified-to-original transfer is presumed. Existence and non-vacuity of compatible paths are part of the hypothesis.

Then, under hypotheses (H2.1)–(H2.3), $B \notin \text{ERViab}_{\mathcal{I}}(\mathcal{V})$: every compatible state may be individually robustly viable, and the belief is nevertheless nonviable, because the state-specific safe actions are incompatible.

Proof. Fix a policy and its held first command a . If $a \notin U^B(B)$, it is inadmissible at a compatible state. Otherwise $a \in U^B(B)$; emptiness of the common active-gradient response set implies some compatible boundary state has an active constraint with a strictly negative drift for this action. Hypothesis (H2.3) supplies an *actual* compatible original-system path from such a state with $q_{j_a}(x_a(t)) < 0$ at arbitrarily small positive times, hence before the next review. The policy therefore fails. This proof is conditional on (H2.3) and the declared held-action class, not a statement about an unrestricted switching policy.

For polyhedral data the emptiness certificate is a Farkas pair — a finite, independently checkable certificate of epistemic infeasibility — and for infinite beliefs the Helly-type witness of Proposition 4 supplies a finite-witness reduction under the convexity hypotheses of Section 3.4; the complete discussion is given in the Supplementary Material (S1).

Remark 1 (numerical robustness of the polyhedral certificate). Linearity makes the Farkas certificate robust in two practical senses. First, bounded data perturbation degrades the margin gracefully: if $\mu \geq 0$ certifies $\mu^\top A = 0$ and $\mu^\top b \leq -\mu_0 < 0$, then for perturbed right-hand sides $b + \Delta b$ with $\|\Delta b\|_\infty \leq \epsilon$ the same multipliers give $\mu^\top (b + \Delta b) \leq -\mu_0 + \|\mu\|_1 \epsilon$, still negative while $\epsilon < \mu_0 / \|\mu\|_1$; the scale-invariant ratio $\mu_0 / \|\mu\|_1$ is a guaranteed perturbation tolerance for this fixed multiplier. Scaling to $\|\mu\|_1 = 1$ does not maximize it; optimizing over admissible multiplier directions is a separate problem. Second, verified approximate data suffice: if the multipliers are computed with residual $r := \hat{\mu}^\top \hat{A}$ and objective $c := \hat{\mu}^\top \hat{b}$,

then, provided $\hat{\mu} \geq 0$ and the row-error bounds are valid, feasibility of some $u \in U$ would imply $-h_U(-r) \leq r^\top u \leq c$, where $h_U(g) := \sup_{v \in U} g^\top v$ is the support function of the action set; hence the test $c + h_U(-r) < 0$ — with all integration, rounding, and enclosure errors charged into c and r before it is applied — certifies infeasibility without exact stationarity. Both extensions share the orientation rule for nonviability certificates: uncertain row data must be relaxed toward a larger feasible controller set, never toward a smaller one; tightening $\hat{b}$ on the strength of an unverified worst case can manufacture a false obstruction.

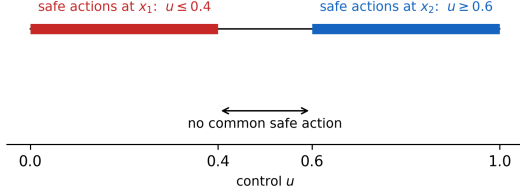


Figure 1: Incompatible safe controls. The states x_1 and x_2 share an observation, but their safe-action sets $[0, 0.4]$ and $[0.6, 1]$ are disjoint, so no common safe action exists (Theorem 2).

The tube-safety form. For a post-observation cell B and a policy required to hold one command over $[0, \Delta]$, let

$$\mathcal{A}_{\text{tube}}(B, \Delta) = \{a \in U^B(B) : \text{every compatible original-system}$$

The quantifier includes all compatible initial states, admissible disturbances and solutions. If a path cannot be continued for the required interval, the model must specify whether this counts as failure rather than treating an empty solution set as safe. Empty tube safety rules out a viable *first action in the declared hold class*; no conclusion follows for unrestricted within-review switching. Instantaneous gradient tests imply this tube obstruction only under the original-system path premise (H2.3).

Proposition 2 (uniform margin gives a tube obstruction at every review length). Let $B \subseteq \mathcal{V}$ be a compact information set with $U^B(B) \neq \emptyset$, f and ∇q_j continuous, and suppose there is $\eta > 0$ such that for every $a \in U^B(B)$ there exist a boundary state $x_a \in B \cap \partial \mathcal{V}$, an active constraint j with $q_j(x_a) = 0$, and a disturbance $d_a \in D(x_a)$ satisfying

$$\nabla q_j(x_a) \cdot f(x_a, a, d_a) \leq -\eta.$$

Assume in addition that each such candidate action has an admissible original-system exiting path as in (H2.3), for its chosen boundary state. Then the tube-safe action set is empty at every review length,

$$\mathcal{A}_{\text{tube}}(B, \Delta) = \emptyset \quad \forall \Delta > 0,$$

and hence $B \notin \text{ERViab}_{\mathcal{I}}(\mathcal{V})$ under sampled review of any length.

Proof. For each fixed a , the assumed original-system exiting path has $q_j(x_a(t)) < 0$ for every sufficiently small $t > 0$. Thus every positive Δ contains an unsafe point, although the permissible exit interval may depend on a . No uniform lower bound on those intervals is needed. The last implication concerns only policies that must hold the command over that review.

Proposition 3 (obstruction ladder). For the declared positive-duration hold class, assuming (H2.3) at every relevant failed active-gradient test and original-system solutions

through the review, the common response sets of Section 2.4 are nested:

$$\mathcal{A}_{\text{tube}}(B, \Delta) \subseteq \mathcal{R}_{\mathcal{V}}^B(B) \subseteq U^B(B) \quad \forall \Delta > 0,$$

and emptiness descends the ladder:

$$\begin{aligned} U^B(B) = \emptyset &\implies \mathcal{R}_{\mathcal{V}}^B(B) = \emptyset \implies \mathcal{A}_{\text{tube}}(B, \Delta) = \emptyset \\ &\implies B \notin \text{ERViab}_{\mathcal{I}}(\mathcal{V}). \end{aligned}$$

Proof. The second inclusion follows from $\mathcal{R}_{\mathcal{V}}(x) \subseteq U(x)$. For the first, a command outside $\mathcal{R}_{\mathcal{V}}^B(B)$ but inside $U^B(B)$ fails an active-gradient test; (H2.3) supplies an admissible original-system path that exits before every positive review time. A command outside $U^B(B)$ is already excluded by the tube definition. The final emptiness implication applies only to policies obliged to hold the first command for this same Δ ; finite-step predecessor sets are not identified with continuous tubes without a sampled-transition model.

The obstruction ladder’s four rungs — admissibility, common safe-action, tube, and recursive — are itemized and shown strict in general in the Supplementary Material (S1, Figure S1); the dynamic exit certificate of Theorem 5 sits below the ladder, emptying a response set at some compatible state even under full information.

Example 1 (hidden-mode conflict). Let an unobserved parameter satisfy $\theta \in \{-1, +1\}$, with $\dot{z} = \theta u$, $u \in \{-1, +1\}$, and constraint $z \geq 0$. At $z = 0$: if $\theta = +1$, only $u = +1$ is safe; if $\theta = -1$, only $u = -1$ is safe. Both compatible states are individually robustly viable under full information (each admits its safe control), but the information set $B = \{(0, +1), (0, -1)\}$ satisfies $\mathcal{R}_{\mathcal{V}}^B(B) = \{+1\} \cap \{-1\} = \emptyset$, so $B \notin \text{ERViab}_{\mathcal{I}}(\mathcal{V})$. This is a purely informational failure: no stochasticity and no estimation-quality argument is involved. The example is the two-action skeleton of every “the stock is either recovering or collapsing, and the safe policy differs between the two” situation.

3.3 The delayed-information obstruction

Theorem 3 (delayed-information obstruction). Let B_0 be an initial post-observation information cell whose members share a current reading. Suppose the following hypotheses hold.

(H3.1) Fix a blind interval $[0, T_{\text{obs}})$ with $T_{\text{obs}} > 0$. For each implementable control history in the declared class, *every* compatible realization while safe produces the same observation record on this interval (up to its first exit). Thus no branch-dependent observation can change the policy’s control before exit; the set-membership information state may evolve, but the common record contains no branch-discriminating signal.

(H3.2) Fix the policy class being certified: by default all implementable measurable blind-window controls; a declared whole-window hold class yields only class-relative nonviability. There is a function q of class C^1 , defined on a neighbourhood of the relevant trajectories, with $\mathcal{V} \subseteq \{q \geq 0\}$, and $\varepsilon > 0$. Let $B_0 \subseteq \mathcal{V}$ be compact, and fix $a > \max_{B_0} q \geq 0$. **For every implementable blind-window control** $u(\cdot)$ in the declared class, measurable and common to all compatible states while there is no informative observation, there exist a minimizer $x_u \in B_0$ of q and an admissible disturbance realization with an absolutely continuous solution from x_u . This solution exists through any first contact with $q = 0$ until either it reaches $q < 0$ or T_{obs} , and follows the common record of (H3.1) until exit. For almost every pre-exit time in the closed strip $0 \leq q(x(t)) \leq a$, *including boundary contact*, its actual derivative obeys

$$\frac{d}{dt} q(x(t)) = \nabla q(x(t)) \cdot f(x(t), u(t), d(t)) \leq -\varepsilon. \quad (2)$$

The realization is chosen *after* u , not before it. This condition is on a genuine admissible trajectory of the original system, not

solely on a convexified disturbance inclusion; no exit is assumed in the hypothesis. It may fail even when a positive-side drift estimate holds.

(H3.3) The timing bound holds:

$$T_{\text{obs}} > \frac{\inf_{x \in B_0} q(x)}{\varepsilon}. \quad (3)$$

Then, under hypotheses (H3.1)–(H3.3), no policy *in the fixed class* is viable from B_0 . When the fixed class is the unrestricted observation-based class of Definition 1, this says $B_0 \notin \text{ERViab}_{\mathcal{I}}(\mathcal{V})$. For a proper declared institutional subclass, it asserts only class-relative nonviability; it does not exclude unrestricted viability.

Proof. Fix a policy in the declared class. Hypothesis (H3.1) gives it a common blind control u ; an action inadmissible on a compatible branch already fails. Otherwise (H3.2) supplies a compatible minimizer and a genuine adverse trajectory. Put $h(t) = q(x(t))$, $h(0) = m = \min_{B_0} q \geq 0$, and $t_* = m/\varepsilon < T_{\text{obs}}$. Suppose there is no strict violation by some $t_1 \in (t_*, T_{\text{obs}})$. As long as $0 \leq h \leq a$, (2) gives $h(t) \leq m - \varepsilon t$. The trajectory cannot first leave this strip through $h = a$ because its derivative is strictly negative there almost everywhere; at a contact $h = 0$ it cannot remain nonnegative through t_1 for the same reason. Hence nonviolation through t_1 would give $0 \leq h(t_1) \leq m - \varepsilon t_1 < 0$, a contradiction. This includes $m = 0$: negative drift is required at the boundary, not a vacuous condition on $h > 0$. The realized minimizer path continues through first contact by (H3.2); contact alone would not imply nonviability for the closed safe set. The theorem need not detect a failure caused only by a non-minimizer branch. so some $t < t_1 < T_{\text{obs}}$ has $q(x(t)) < 0$, contrary to safety. The policy was arbitrary; the quantifier is $\forall \pi \exists (x_u, d_u)$, not a demand that every trajectory drift downward.

Hypothesis (2) is a condition on the data of the problem, not on the policies, and condition (3) is a sufficient timing condition for this obstruction; evading it does not itself establish viability. The complete discussion of both hypotheses, and the formal-zero-margin comparison with Theorem 2, are given in the Supplementary Material (S1).

Remark 2 (threshold form of the timing obstruction). The delayed-information obstruction has a sharp threshold. For a fixed declared blind-window class Π , let $\sigma_{\Pi}^*(B_0)$ be its guaranteed survival time: the largest duration such that some control in Π keeps every observation-equivalent branch in $\mathcal{V}$ until then,

$$\sigma_{\Pi}^*(B_0) = \sup_{u(\cdot) \in \Pi} \inf_{x_0 \in B_0, d(\cdot)} \tau(x_0, u, d),$$

with the infimum over branches that remain observation-equivalent on $[0, T_{\text{obs}}]$ and τ the first violation time under the convention of Theorem 3. Then

$$\sigma_{\Pi}^*(B_0) < T_{\text{obs}} \implies \text{no policy in } \Pi \text{ is viable from } B_0.$$

For the unrestricted observation-based class this implies $B_0 \notin \text{ERViab}_{\mathcal{I}}(\mathcal{V})$, but for a proper hold class it does not. Before T_{obs} , a policy cannot condition its action on which branch realizes. Condition (3) certifies the class-specific threshold: under (H3.1)–(H3.2), *for every control in Π , at least one compatible branch* exits by time $\inf_{x \in B_0} q(x)/\varepsilon$, hence $\sigma_{\Pi}^*(B_0) \leq \inf_{x \in B_0} q(x)/\varepsilon$, and (3) states that this upper bound lies below T_{obs} ; the threshold itself may be strictly smaller, in which case nonviability holds even when (3) fails. The threshold is the blind-window response correspondence of the timing layer: its emptiness is the timing certificate, the delayed-information analogue of the tube and common-action correspondences.

Checkability of the timing certificate. The drift hypothesis (H3.2) is finitely checkable whenever the blind-window

control class is finite: if B_0 is finite and the admissible blind-window controls form a finite family (hold-until- T_{obs} controls with u in a finite set, or piecewise-constant controls on a finite partition), the strengthened certificate requires, for each candidate, an admissible adverse solution with the closed-strip derivative condition through first boundary contact; finite enumeration of controls alone does not establish that continuous-time realization. The hidden-regime instance of Section 9 is a *discrete analogue* of this check for its declared whole-window hold class, not an application of the continuous drift theorem: the worst branch declines at unit rate. For integer blind steps the maximal safe count is $\sigma_{\text{hold}}^*(B_0) = \lfloor z_0 - 1 \rfloor$, so the hold-class certificate fires exactly when the integer $T_{\text{obs}} > z_0 - 1$. For per-step switching this calculation does not apply; in particular alternating actions can save both branches from $z_0 \geq 2$. For continuous control classes the check remains a template, part of Open Problem 1; over polytope-declared classes, however, the check is exact and finite (Theorem 4), and the class declaration itself is load-bearing (Example 2).

Theorem 4 (linear-programming instantiation of the timing certificate). Let the blind window have K discrete steps, finitely many branches $j \in J$ with robust affine dynamics $x^+ = A_j x + B_j u + E_j w$, $w \in W_j$ a polytope, known initial branch states $x_{j,0}$, polyhedral safe set $\mathcal{V} = \{x : Fx \leq g\}$, and declared blind-window class $\Pi_B = \{(u_0, \dots, u_{K-1}) : u_k \in U\}$ for a polytope U . Then: (i) $\sigma^*(B_0; \Pi_B)$ is computed exactly by at most $\lceil \log_2 K \rceil + 1$ linear programs of size polynomial in K , $|J|$, and the description lengths of U , W_j , $\mathcal{V}$ — unrolling the dynamics makes each robust floor row one linear inequality in the control sequence, the disturbance entering only through precomputed support constants — and with rational data every step is exact. (ii) When the per-step scenario choices are independent, infeasibility at horizon K is certified by Farkas multipliers: nonnegative weights on the rows whose aggregate control projection vanishes and whose aggregate constant is negative — the timing certificate in adjoint-row form. (iii) The program’s feasible set is the declared class itself, so the computation is sound and complete over Π_B : no relaxation gap arises. (iv) The declaration is load-bearing: a polytope-per-step program may be an outer relaxation of a *separately declared whole-window hold class* $\Pi_{\text{hold}} = \{(u, \dots, u) : u \in \{-1, +1\}\}$, not of a class allowing arbitrary per-step switching with the same two actions. For the non-decaying hidden-regime branches $z^+ = z \pm u$ at $z_0 = 2$ and floor $z \geq 1$, either held action survives only one blind step, whereas the relaxation $U = [-1, 1]$ admits $u \equiv 0$ through every horizon. The gap grows without bound as K increases (the relaxed infinite-horizon survival time is infinite). Switching $+1, -1, +1, \dots$ is allowed in the *per-step* finite-action class and rescues it; the asserted gap is specifically between the whole-window hold class and its polytope-per-step relaxation.

Proof sketch. For branch j , horizon step k , floor row r , and independent disturbances $w_{j,t} \in W_j$, the robust inequality is

$$\sum_{t=0}^{k-1} F_r A_j^{k-1-t} B_j u_t \leq g_r - F_r A_j^k x_{j,0} - \sum_{t=0}^{k-1} h_{W_j} \left(E_j^\top (A_j^{k-1-t})^\top F_r^\top \right).$$

Unrolling the affine dynamics makes each branch state $x_{j,k}$ affine in $(u_0, \dots, u_{k-1})$, so every robust floor row is one linear inequality once $h_{W_j}(E_j^\top (A_j^{k-1-t})^\top F_r^\top)$ is precomputed; this row assumes independent per-step disturbances $w_{j,t} \in W_j$ and floor row F_r for branch j , step k , and disturbance time t . For coupled disturbance scenarios the joint support must be used instead. Feasibility of the horizon- K' system is monotone in K' (prefix truncation), giving (i) by bisection, and Farkas’ lemma gives (ii). Claim (iii) is definitional: the program’s variables are the declared controls themselves. Claim (iv) is the computation displayed in Example 2; the complete proof, including the duality computation, is in Supplementary S2.

Example 2 (decaying LP threshold and a separate held-class relaxation gap). Two branches share $z^+ = \frac{9}{10}z \pm u$ from a common z_0 , floor $\mathcal{V} = \{z \geq 1\}$, declared class $U = [-1, 1]$ per step. The branch sum $z_k^+ + z_k^- = 2(\frac{9}{10})^k z_0$ is invariant under the control, so both floors force $z_0 \geq (\frac{10}{9})^k$ at every step k : the window- K program is feasible exactly when $z_0 \geq (\frac{10}{9})^K$. The level- k Farkas certificate is the pair of floor rows summed — their control coefficients cancel pairwise — leaving $2(\frac{9}{10})^k z_0 \geq 2$; and $u \equiv 0$ attains feasibility on the boundary $z_0 = (\frac{10}{9})^K$, keeping the branches balanced at $(\frac{10}{9})^{K-k} \geq 1$. The program is thus a complete characterization of this decaying polytope-declared class. The *different* non-decaying hidden-regime example used for the unbounded gap in clause (iv) has $z^+ = z \pm u$, $z_0 = 2$, and a whole-window finite hold class: either hold survives one step, while the relaxed action $u = 0$ survives arbitrarily many. Its gap is not a survival-time formula for the decaying system above. Nor does it apply to the per-step finite class, whose alternating actions are allowed; the two computations are separated in Supplementary S1.

Remark 3 (adjoint-row correspondence). The Farkas multipliers of (ii) are the finite analogue of the adjoint safety rows of the continuous-to-finite bridge: a weight $\lambda_{j,k,i}$ on the i -th floor row of branch j at step k plays the role of an adjoint variable on a continuous-time safety row, the vanishing aggregate control projection is the pooled adjoint condition, and the negative aggregate constant is the certificate margin. The control coefficients of Example 2 cancel pairwise within each level, which is the discrete image of the pooled adjoint rows' cancellation. The bridge to continuous time — error-controlled sampling of these programs — is developed in a companion work.

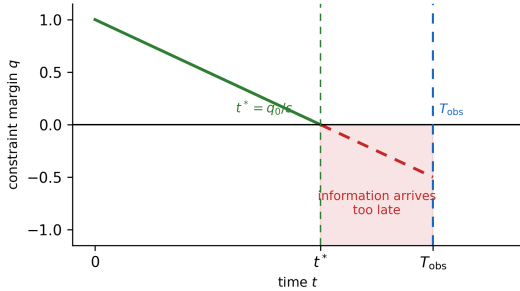


Figure 2: Delayed information. The constraint margin q reaches zero at $t^* = q_0/\varepsilon$, before the first informative observation at T_{obs} : the information is accurate but arrives too late (Theorem 3).

3.4 The sparse finite witness

Proposition 4 (sparse common-action witness). Let $B \subseteq \mathcal{V}$ be an information set, $U(x) \subseteq U$ with $U \subseteq \mathbb{R}^m$ compact and convex and $U(x)$ convex and closed, let the dynamics be affine in the control, $f(x, u, d) = f_0(x, d) + f_u(x, d)u$, with $D(x)$ compact, and let $\mathcal{V}$ have C^1 constraint functions q_j . Then each safe-control set $\mathcal{R}_{\mathcal{V}}(x)$ is a compact convex subset of U , and the safety-case common-action obstruction $\bigcap_{x \in B} \mathcal{R}_{\mathcal{V}}(x) = \emptyset$ holds if and only if it is witnessed by at most $m+1$ compatible states: there exist $1 \leq \ell \leq m+1$ distinct states $x_1, \dots, x_\ell \in B$ with $\bigcap_{i=1}^\ell \mathcal{R}_{\mathcal{V}}(x_i) = \emptyset$. The admissibility analogue (with $\mathcal{R}_{\mathcal{V}}(x)$ replaced by $U(x)$) holds under the same convexity assumptions.

Proof. For fixed (x, j, d) the inequality $\nabla q_j(x) \cdot f(x, u, d) \geq 0$ is affine in u , so $\mathcal{R}_{\mathcal{V}}(x)$, an intersection of closed halfspaces (over active j and $d \in D(x)$) with the closed convex $U(x)$, is closed and convex; contained in the compact U , it is compact.

If the intersection over all $x \in B$ is empty, then by compactness some finite subfamily of $\{\mathcal{R}_{\mathcal{V}}(x)\}_{x \in B}$ already has empty intersection, and by the finite Helly theorem at most $m+1$ members suffice; the converse direction is immediate. $\square$

Scope. Both conclusions rest on convexity: the finite Helly theorem is what bounds the witness by $m+1$, and beyond convex safe-control sets neither the bound nor the finite checkability of the certificate survives in general.

A worked certificate (two floors, one scalar action). Two floors demand incompatible extraction rates on a single control $u \in [0, 1]$. At the compatible state x_1 (floor 1 active) the safe controls are $u \leq 0.4$; at x_2 (floor 2 active) they are $u \geq 0.6$. The observation merges x_1 and x_2 , so a common safe action must satisfy both. Writing the constraints as $Au \leq b$ with $A = (1, -1)^\top$ and $b = (0.4, -0.6)^\top$, the stacked system $u \leq 0.4, -u \leq -0.6$ is infeasible; the Farkas certificate is $\lambda = (1/2, 1/2)$ (normalized, $\|\lambda\|_1 = 1$), for which $\lambda^\top A = 0$ and $\lambda^\top b = -0.1 < 0$. Under the normalization $\|\lambda\|_1 = 1$, the magnitude $|\lambda^\top b| = 0.1$ is the infeasibility margin of the stacked system, shrinking to zero exactly as the two safe-action sets come to touch. The multiplier exhibits the contradiction as a convex combination of the two floor constraints: no common extraction rate respects both floors, and any observation that separates x_1 from x_2 restores one-step viability (Figure 1). With $m = 1$, the two incompatible states are exactly the $m+1$ -state witness of Proposition 4.

Scope of the convexity hypotheses. Convexity is essential to the witness: without it the bound fails even in a single control dimension. Take $U = \mathbb{R}$ ($m = 1$) and three nonconvex safe-action sets $\mathcal{R}_1 = \{0, 1\}$, $\mathcal{R}_2 = \{0, 2\}$, $\mathcal{R}_3 = \{1, 2\}$. Each pair intersects ($\mathcal{R}_1 \cap \mathcal{R}_2 = \{0\}$, $\mathcal{R}_1 \cap \mathcal{R}_3 = \{1\}$, $\mathcal{R}_2 \cap \mathcal{R}_3 = \{2\}$), yet the three-way intersection is empty, so no $(m+1) = 2$ -state witness exists. The convexity hypotheses of Proposition 4 exclude exactly this failure mode.

Scope of the count: instantaneous actions, not window-policies. The $m+1$ bound counts independent instantaneous decisions: it applies to a common action $u \in U \subseteq \mathbb{R}^m$, and it fails for common blind control *functions*. Partition a blind window into Q blocks $I_1, \dots, I_Q$ and let $q+1$ branches be indistinguishable throughout, with scalar input ($m = 1$): branch $j \leq q$ evolves as $\dot{x}_j = \mathbf{1}_{I_j}(t)(u - 1)$ from $x_j(0) = 0$ under the floor $x_j \geq 0$, which forces $u \equiv 1$ on every block (each $w_j = \int_{I_j} u dt = 1$), while branch $q+1$ evolves as $\dot{x}_{q+1} = -u$ from $x_{q+1}(0) = q - \varepsilon$ under the same floor, forcing $\sum_j w_j \leq q - \varepsilon$. Summing the forced withdrawals gives $0 \leq -\varepsilon$: the full system is infeasible, while removing any single branch restores feasibility ($u \equiv 1$, or $u = 0$ on the dropped branch's block and 1 elsewhere), so every obstruction involves all $q+1$ branches — far more than $m+1 = 2$. The adjoint kernels here are the block indicators, of rank q : for window-policies the governing count is the information-time rank, and sparsity claims for review-interval and delayed-sensing certificates must be stated in the block dimension mQ , not the instantaneous dimension m (cf. the block structure of the delayed-information check in Section 8).

3.5 The finite-time exit certificate

The base obstruction operates before any observation-theoretic argument: it certifies membership in the full-information “shadow” — the complement of the viability kernel (Aubin, 2001; Aubin and Catté, 2002). It is a drift condition under which *no* control — of any information structure — can be viable. Put in management terms, when the dynamics themselves force exit before any control can react, no monitoring design can rescue viability, and the certification problem is closed without observation-theoretic reasoning.

Theorem 5 (finite-time exit certificate). Let $q : X \rightarrow \mathbb{R}$ be a constraint function of class C^1 — the class assumed throughout the exit theorems — with $\mathcal{V} \subseteq \{q \geq 0\}$, and suppose the following hypotheses hold.

(H5.1) There exist constants $a > 0$ and $\varepsilon > 0$ such that on the strip $\mathcal{S}_a = \{x : 0 \leq q(x) \leq a\}$,

$$\sup_{u \in U(x)} \inf_{d \in D(x)} D^+ q(x; f(x, u, d)) \leq -\varepsilon \quad \forall x \in \mathcal{S}_a, \quad (4)$$

where $D^+ q(x; v) = \limsup_{h \downarrow 0} [q(x + hv) - q(x)]/h$ is the upper right Dini derivative of q in direction v (for C^1 q , $D^+ q(x; v) = \nabla q(x) \cdot v$ — the form the proof uses).

(H5.2) **Original-system adverse realization for the declared control class.** For every admissible control history in the stated class and every $x_0 \in \mathcal{S}_a$, there exists a compatible absolutely continuous original-system trajectory $x(\cdot)$ and measurable disturbance $d(\cdot)$, with $d(t) \in D(x(t))$ and $\dot{x} = f(x, u(t), d(t))$ almost everywhere. It is defined beyond $q(x_0)/\varepsilon$ unless it has already reached $q < 0$, and while $0 \leq q(x(t)) \leq a$ before exit it satisfies $\frac{d}{dt} q(x(t)) \leq -\varepsilon$ almost everywhere. The trajectory continues through first contact long enough to decide strict exit. This is a substantive hypothesis about the *original* game, not a consequence of ambient lower semicontinuity or closed graph of D alone. For an open-loop control one admissible path may depend on the selected control; disturbance-reactive or merely measurable closed-loop policies require their own compatible-solution premise.

Then, under (H5.1)–(H5.2), for every control in the *declared* class and $x_0 \in \mathcal{S}_a$ an admissible original-system trajectory strictly violates $q \geq 0$ before every time strictly greater than $q(x_0)/\varepsilon$. The infimum of strict-violation times is at most $q(x_0)/\varepsilon \leq a/\varepsilon$; a strict violation *at* that bound is not asserted. Since $\mathcal{V} \subseteq \{q \geq 0\}$, the trajectory defeats viability in this class. At $q = 0$ a state may still be safe.

Proof. Fix a control and the compatible pair of (H5.2). Let $h(t) = q(x(t))$, $t_0 = h(0)/\varepsilon$. If the pair stays safe through a time $t_1 > t_0$ before its guaranteed continuation ends, it cannot first cross upward through $h = a$, and on the still-safe portion $0 \leq h \leq a$ integration gives $0 \leq h(t_1) \leq h(0) - \varepsilon t_1 < 0$, a contradiction. Therefore it strictly exits before any such t_1 ; a single strict violation so obtained precedes every later deadline. The argument never integrates the drift condition on a path after it has left its stated strip. Existence of the compatible original pair, not a closed-graph selection slogan or an unproved relaxation transfer, is the load-bearing premise.

Remark 4 (comparison-function form). If a state-dependent decline bound $\dot{q} \leq -\alpha(q)$, $\alpha : (0, a] \rightarrow (0, \infty)$, holds along a compatible adverse original trajectory while $q > 0$, and $\int_0^{q(x_0)} ds/\alpha(s) < \infty$, comparison bounds the time of *first boundary contact*. It does not alone give strict exit from the closed set $\{q \geq 0\}$: the solution of $\dot{q} = -\sqrt{q}$ from $q(0) = 1$ reaches zero at time two and can stay there. Strict exit requires a compatible continuation with negative drift at or through contact while the path remains safe. A value assigned to $\alpha(0)$ without such a continuation is not a substitute for this hypothesis.

The condition (4) has a control-before-disturbance, pointwise reading. A sound pathwise certificate still requires (H5.2) in the original system for the declared policy class. In particular, an open-loop adverse solution is not automatically a solution against a controller observing the current disturbance; a common admissible disturbance d^* with $\sup_{u \in U(x)} \nabla q(x) \cdot f(x, u, d^*) \leq -\varepsilon$ throughout the strip and compatible closed-loop solutions is one sufficient additional condition for such a disturbance-reactive class. A pointwise minimax interchange, even where justified, does not supply one state-uniform d^* .

The certificate is information-independent in its drift inequality, *not* unconditional in solution existence or policy class. A state-dependent barrier may be used with output feedback; only an explicitly observation-only barrier subclass must be fibre-constant.

3.6 Epistemic emptiness by admissibility: a minimal construction

The admissibility obstruction applies to systems in which every state is individually viable under full information and the kernel empties only under the observation structure. The minimal construction of this section empties the kernel through *admissibility* — a state-dependent control set whose common action is forced and unsafe at the boundary; the common-action obstruction of Section 3.2 and Example 1’s hidden-parameter conflict are the informational cases proper, merging states with incompatible safe controls. In ecological terms, each stock is manageable on its own; the failure arises because the indicator cannot tell the manager which stock is present.

Proposition 5 (epistemic emptiness by admissibility — a minimal construction). There exists a system with $\text{Viab}(\mathcal{V}; U, \pi_{\text{perf}}) = \mathcal{V} \neq \emptyset$ such that, for a non-injective observation map O , every *full-fibre* initial information state — the observation fibres $B_0 = O^{-1}(O(S_0))$, $S_0 \in \mathcal{V}$ — lies outside $\text{ERViab}_{\mathcal{I}}(\mathcal{V})$: the epistemic kernel over the fibre-induced initial beliefs is empty. The mechanism is *admissibility* rather than a hidden mode: the state-dependent control set of the construction leaves a single common action at the merged belief — one that is outward-drifting at the boundary state — so the emptiness is the common-action obstruction (Theorem 2’s safety case) at minimal size, produced by the control-set primitive rather than by information; the hidden-mode instance is Example 1.

Proof sketch (construction). Take $X = [1, 2]$ with dynamics $\dot{S} = u - r(S)$, and choose $r(S) = S$ (continuous, injective, and bounded below by 1 on $[1, 2]$). Set the *unrelaxed, nonconvex* $U(S) = \{0, S\}$, $\mathcal{V} = [1, 2]$, and the constant observation $O \equiv 0$. Under full information $u = r(S)$ gives $\dot{S} = 0$, so every state is viable. Under $O \equiv 0$ the common control set over the fibre is $\{0\}$ — the only action shared by all $U(S)$ is zero — so every observation-based policy must hold $u = 0$. Then $S(t) = S(0)e^{-t}$ crosses the lower boundary at $t = \log S(0) \leq \log 2$ for $S(0) > 1$, and leaves immediately from $S(0) = 1$.

The construction is the minimal instance of Theorem 2’s safety obstruction, with a nonconvex state-dependent admissibility set — with a constant control set in the same plant the fibre is epistemically viable — so the empty-kernel phenomenon is epistemic rather than dynamic; the complete isolation of the mechanism is given in the Supplementary Material (S1). Convexifying the *controller* sets to $\text{co} U(S) = [0, S]$ is a different problem: then $u \equiv 1$ is common and keeps every state in $[1, 2]$; no such controller convexification is used in the original construction.

3.7 A recourse certificate for the post-observation mode

The certificates of the preceding subsections fire before information discriminates: the common-action obstruction is instantaneous (Section 3.2) and the timing obstruction exits within the blind window (Section 3.3, Remark 2). Section 6.5(ii) isolates the complementary mode: insufficient post-observation recourse. In that mode a common blind-window control keeps every branch safe, yet no observation-adapted control can recover every branch once the distinguishing observation arrives; the discrete instance of Supplementary A.3 exhibits the mode while no certificate fires. This subsection

supplies a continuous-time certificate for the mode under an affine-state, affine-input plant with affine safety facets and exogenous disturbances. Its soundness covers every measurable information-adapted policy *within that plant class*. Completeness is not claimed, because the certificate relaxes the post-observation class upward: an oracle grants each stored branch its own controller whether or not the sensor reveals that much, and an oracle failure is only a sufficient condition for real failure.

Proposition 6 (oracle-recourse obstruction). Assume the controlled dynamics are affine in the *state and input* for each branch and fixed disturbance scenario: $\dot{x}_\theta = A_\theta^d(t)x_\theta + g_\theta^d(t) + B_\theta^d(t)u$, where the coefficients are bounded and measurable on $[0, T]$ and, once the scenario is fixed, do not depend on the controlled trajectory or control. Disturbance scenarios are exogenous: each admissible $d(\cdot)$ remains admissible under every policy. Let the safety facets be affine, $q_j(x) = c_j^\top x + b_j$, with safety $q_j \geq 0$, and let $U \subseteq \mathbb{R}^m$ be compact and convex. Let branches $x_1, \dots, x_L \in B_0$ be observation-equivalent on $[0, \tau]$ with horizon $T > \tau$. Fix a label $\ell = (\theta, j, t)$, $t \in (\tau, T]$, and a fixed admissible scenario $d(\cdot)$. Let $\bar{x}_\theta^d$ be its zero-control trajectory, $X_\theta^d(t, \sigma)$ the transition matrix of A_θ^d . For each stored label ℓ , write $B_\ell^d(\sigma) := B_{\theta_\ell}^d(\sigma)$, so labels from different branches use their own input coefficients. The state-independent input coefficient and affine facet are load-bearing: a general state-dependent $B(x)$ or nonlinear facet requires a validated one-sided remainder before the displayed certificate can be used. Define the input kernel

$$k_\ell^d(\sigma) = \mathbf{1}_{\{\sigma \leq t\}} (B_\ell^d(\sigma))^\top (X_\theta^d(t, \sigma))^\top \nabla q_j(\bar{x}_\theta^d(t)),$$

let $\beta_\ell^d = q_j(\bar{x}_\theta^d(t))$ denote the zero-control facet margin at the label, and write $\phi_U(g) := \min_{v \in U} g^\top v$. For weights $\lambda_\ell \geq 0$, $\sum_\ell \lambda_\ell = 1$, define the recourse certificate

$$\Gamma_h(\lambda) = \inf_{d(\cdot)} \left\{ \sum_\ell \lambda_\ell \beta_\ell^d + \int_0^\tau h_U \left(\sum_\ell \lambda_\ell k_\ell^d(\sigma) \right) d\sigma + \sum_\ell \int_\tau^T h_U (\lambda_\ell k_\ell^d(\sigma)) d\sigma \right\},$$

with $h_U(g) := \max_{v \in U} g^\top v$ the support function of U ; the infimum runs over admissible exogenous disturbance scenarios. Variation of constants evaluates a bracket for a fixed scenario; it does not by itself compute the scenario infimum. An admissible scenario with a strictly negative bracket (an upper bound on the infimum) already supplies a failure witness, whereas a negative lower bound alone does not. If $\Gamma_h(\lambda) < 0$, then $B_0 \notin \text{ERViab}_\mathcal{I}(\mathcal{V})$: for every admissible policy, at least one compatible stored branch violates a facet under the chosen exogenous scenario (the violating branch may depend on the policy).

Proof sketch. If the infimum defining Γ_h is negative, choose one exogenous admissible scenario whose bracket is negative; it remains admissible against every policy. For that fixed scenario, variation of constants for the stated affine-state system and affine facets expands the weighted facet sum *exactly* into the β -term, the window pairing (one common measurable signal on $[0, \tau]$, by the blind-window reduction of Theorem 3), and the branch pairings (separate post-reveal signals); each pairing is bounded above by its support value, so the weighted facet sum is strictly negative. A strictly negative weighted sum forces some facet value strictly negative — the facet is violated along its branch — and the window signal was arbitrary, so every policy in the class fails. The complete proof, the comparison with the attainment-special ϕ_U -variant, and the position of the pooling weights are given in the Supplementary Material (S1). $\square$

Example 3 (three branches, delayed recourse). A planar braking instance in which the obstruction is purely one of post-observation recourse. Branches $j = 1, 2, 3$ run $\dot{p} = v$, $\dot{v} = u$ (no disturbance) from $p_j(0) = \frac{34}{25}n_j$, $v_j(0) = n_j$, where $n_1 = (1, 0)$, $n_2 = (-\frac{3}{5}, \frac{4}{5})$, $n_3 = (-\frac{3}{5}, -\frac{4}{5})$; the actuator set is $U = \text{conv}\{\pm n_1, \pm n_2, \pm n_3\}$ (the hexagon $|u_2| \leq \frac{4}{5}$, $|2u_1 + u_2| \leq 2$); safety is $n_i^\top p \leq 2$ for $i = 1, 2, 3$ with $\|v\|_\infty \leq \frac{6}{5}$; the realized branch is revealed at time τ . Every branch is viable under full information — brake along $-n_j$ for one second; peak critical position $\frac{34}{25} + \frac{1}{2} = \frac{93}{50} < 2$ — and $u \equiv 0$ keeps every branch strictly safe through any blind window of length up to $\frac{16}{25}$ ($n_j^\top p_j(\tau) = \frac{34}{25} + \tau$): no instantaneous or blind-window certificate fires. Yet the label $(j, n_j^\top p \leq 2, \tau + 1)$ along branch j has zero-control margin $\beta_j = \frac{16}{25} - \tau - 1$ and kernel $\mathbf{1}_{\{\sigma \leq \tau+1\}}(\tau + 1 - \sigma)n_j$; the weights $\lambda = (\frac{3}{8}, \frac{5}{16}, \frac{5}{16})$ pool the pre-observation kernel to zero, and the certificate evaluates to

$$\Gamma_h(\tau) = \left(\frac{16}{25} - \tau - 1 \right) + \sum_j \lambda_j \int_\tau^{\tau+1} (\tau + 1 - \sigma) d\sigma = \frac{7}{50} - \tau,$$

the pooled pre-observation kernel vanishing under λ and each post-reveal branch's pairing bounded by its support value ($\sum_j \lambda_j/2 = \frac{1}{2}$). For $\tau > \frac{7}{50}$ the certificate fires and the triple is nonviable — the blind window accumulates braking liability that the post-observation authority cannot discharge on every branch at once — while the hold-then-brake policy ($u = 0$, then $-n_j$ for one second) is safe exactly to $\tau = \frac{7}{50} = 0.14$: the threshold is exact. The obstruction is genuinely three-way: at $\tau = \frac{1}{5}$, every pair of branches is jointly viable under a common blind control (worst critical positions $\frac{2469}{1250} = 1.9752$ and $\frac{2419}{1250} = 1.9352$, both below 2), so six of the seven nonempty priors over these branches are viable at a delay where the triple is not. The discrete analogue in which no certificate fires is Supplementary A.3; here the certificate fires, $\Gamma_h(\frac{1}{5}) = -\frac{3}{50} < 0$.

The certificate is deliberately asymmetric, like the calculus it joins: it is sound against every measurable policy of the information class, its stored labels are finite and independently checkable (exact arithmetic throughout the instance above), and its incompleteness is explicit — a complete continuous-time characterization of the recourse mode remains open (Section 6.5(ii)).

3.8 A worked case: a static aggregate LP with a rational dual

This subsection gives a simple static aggregate-allocation LP verified in exact rational arithmetic. Because its constraints depend only on Y , it does not demonstrate a nontrivial state-varying-fibre Helly witness or an infinite-dimensional dynamic calculation.

The system. A state is a pair of stocks $(x_1, x_2) \geq 0$ carrying a demand floor on each coordinate. Let $Y = x_1 + x_2$ be the aggregate, and let the Y -fibre be the set of all states with that aggregate — a continuum of states, since (x_1, x_2) ranges over the whole segment $x_1 + x_2 = Y$, $x_1, x_2 \geq 0$. The control set is $U = \{u \geq 0 : u_1 + u_2 \geq 2\}$, the allocations meeting total demand. Feasibility on the fibre is constrained by Y -dependent caps

$$\text{cap}_1(Y) = \frac{3}{2} - \frac{Y-2}{10}, \quad \text{cap}_2(Y) = \frac{59}{50} - \frac{Y-2}{10},$$

an allocation $u \in U$ being feasible on the fibre exactly when $u_i \leq \text{cap}_i(Y)$ for each i . Fibre feasibility — not the transition dynamics — is the audited object here, and the scope of the claim below is correspondingly static.

The certificate. Viability of the fibre turns on a single rational inequality. The cap sum is the affine function $\frac{67}{25} - \frac{Y-2}{5}$, which equals 2 at exactly one aggregate:

$$Y^* = \frac{27}{5}.$$

For $Y \leq Y^*$ the fibre is feasible; for $Y > Y^*$ it is not, and the obstruction is certified by the ℓ_1 -normalised dual measure $\lambda = (\frac{1}{2}, \frac{1}{2})$ on the two cap rows, which for every $u \in U$ gives

$$\lambda^\top(\text{cap} - u) \leq \frac{\text{cap}_1(Y) + \text{cap}_2(Y)}{2} - 1 = -\frac{Y - Y^*}{10} < 0.$$

The certificate is therefore two rationals and one inequality, and its margin is exact:

$$\text{margin}(Y) = \frac{Y - \frac{27}{5}}{10}.$$

The two audited fibres. At $Y = 5$ the cap sum is $\frac{52}{25} > 2$ and the fibre is feasible, with witness $u = (\frac{6}{5}, \frac{4}{5})$: it meets demand ($\frac{6}{5} + \frac{4}{5} = 2$) and respects both caps. At $Y = 6$ the cap sum is $\frac{47}{25} < 2$, the fibre is obstructed, and λ certifies it with margin exactly $\frac{3}{50}$. Both statements, and the crossover, were recomputed independently in exact rational arithmetic; the margin identity $\text{margin}(Y) = (Y - \frac{27}{5})/10$ was checked at two hundred aggregates on the half-line $Y \geq Y^*$ with no violation.

What this establishes. The two aggregate caps yield a finite LP, which the rational witness and margin decide for every $Y \geq \frac{27}{5}$. This result does not rely on enumerating the individual points in a fibre.

Scope. This is a static two-variable LP: the caps depend only on Y , so every point of the same continuum fibre imposes the *same* two inequalities. Its exact rational certificate checks this LP and illustrates a shared-action failure, but the continuum does not add a nontrivial finite-witness or Helly problem here. No dynamic epistemic kernel is decided. Window-only data do not determine the post-reveal recourse verdict of Proposition 10; this is not a computability-undecidability claim.

3.9 Mechanized verification of the discrete core

The results above are proved by hand. The earlier discrete formalization covers a separate finite fragment; its recorded build does not prove the continuous-time Theorems 3 or 7, nor validate the finite-state Table 4. That table is checked by a separate exhaustive script. These statements have not been independently reviewed in Lean or by an external referee.

What was checked. The finite-horizon theory, the kernel recursion, the common-action obstruction, the certifier, and the monotonicity statements are formalized in Lean 4 in `Formalizations/P1_Obstruction`, which imports only the project prelude. The project builds cleanly: all 60 build jobs pass. The formalization comprises 58 declarations (33 theorems and 25 definitions, excluding one `structure` and three `inductive` declarations), and the mapping onto the results above is direct:

Checked Lean declarations and their limits. In Lean, an unfinished proof appears as the axiom `sorryAx` in the axiom footprint of every declaration that depends on it. That footprint was inspected for the declarations above. Several depend on no axioms at all (`finite_horizon_sound`, `tree_sound`, `Wk_antitone`,

Lean declaration	Claim supported here
<code>finite_horizon_sound</code>	Finite recursion (Theorem 1)
<code>finite_horizon_complete</code>	Finite recursion (Theorem 1)
<code>Wk_antitone</code>	Finite recursion (Section 3.1)
<code>Wk_descending</code>	Finite recursion (Section 3.1)
<code>commonSafe</code>	Finite one-step result (Theorem 6)
<code>Blocked</code>	Finite one-step result (Theorem 6)
<code>blocked_iff</code>	Finite one-step result (Theorem 6)
<code>preOp_mono</code>	Finite predecessor only; not full Proposition 8

Table 1: Selected Lean declarations. None formalizes the continuous-time Theorems 3 or 7 or independently checks Table 4.

`Wk_descending`, `preOp_mono`, `commonSafe`); the remainder depend only on `Classical.choice`, `propext` and `Quot.sound`, the standard axioms of Lean’s classical logic. `sorryAx` is absent from the checked P1 declarations and the audited aggregate-import footprints; project-wide generated native-evaluation axioms are disclosed in the provenance paragraph below. **Current Lean provenance.** The committed project pin `v4.34.1` was rebuilt in GitHub Actions (default Lake target: 60/60 jobs, exit 0; <https://github.com/MIKEAA2020/general-sustainability/actions/runs/36946730855>). The exact project source and two footprint-check `.lean` files are archived with SHA-256 manifests. A comment-aware scan of all 62 source files finds no `sorry`, `admit`, explicit `axiom` or `constant` declaration. Twenty-one named footprints contain no `sorryAx`; the aggregate-import environment exposes three source-level `native_decide` uses in the P3 support/value examples, yielding nine distinct generated axiom names under this pin. These are executable computational certificates that extend kernel-only trust to the Lean compiler/runtime; they are not admitted gaps and must not be described as zero axioms or as kernel-only proofs. Three preserved historical modules outside the aggregate import closure are not certified by the default build. The pinned declared-exceptions gate passed in GitHub Actions (run 36990482599): only the nine exact generated names from three approved P3 source sites are permitted; any change to those sites or names requires separate review and re-approval. The separate strict zero-exception check exits 2 by design, while the build itself passed 60/60; the two verdicts must not be conflated. The `v4.14.0` comparison of the *current* source fails in the prelude and does not reproduce the old source run.

Two scope limits, stated plainly. First, the formalization covers the *discrete* core. The paper’s continuous-time statements — the Dini-derivative arguments, the exit certificates, the timing bounds — live in an analysis setting that a dependency-free layer does not reach, and they are *not* formalized. The claim is that the discrete core is machine-checked, not that the whole paper is.

Second, completeness is formalized with respect to policy *trees*, not stationary policies: `finite_horizon_complete` returns a `PolicyTree`, a history-dependent object. Where the paper asserts that a stationary policy suffices, that assertion rests on the hand proof, not on the formalization.

Scope of formal support. The named Lean declarations establish the discrete results under their stated hypotheses. Continuous-time exit and Dini-derivative arguments, and any stationary-policy sufficiency statement not encoded as a policy tree, remain supported by their displayed mathematical proofs rather than by these declarations.

4. Certification Limits

4.1 Exact certification and the fibre criterion

The obstruction theorems of Section 3 concern *policies*. Sustainability governance also uses *certificates*: verdicts — safe or unsafe — computed from observations alone, without reference to a policy. The question is when such verdicts can be exact. For a manager operating under partial ecological information, the question is whether an indicator reading alone can ever settle whether the system is on the safe side of a specified floor.

Definition 2 (exact certifier). For an admissible domain $Z \subseteq X$ and a safe set $K \subseteq Z$, an **exact certifier** based on the observation map $O : Z \rightarrow Y$ is a function $C : Y \rightarrow \{0, 1\}$ such that

$$C(O(z)) = 1 \iff z \in K \quad \text{for every } z \in Z.$$

Proposition 7 (observation-fibre criterion). An exact certifier based on O exists if and only if membership in K is constant on every observation fibre:

$$O(z_1) = O(z_2) \implies [z_1 \in K \iff z_2 \in K] \quad \forall z_1, z_2 \in Z, \quad (5)$$

equivalently, $K = O^{-1}(O(K))$ on the admissible domain.

Proof sketch. If C exists and $O(z_1) = O(z_2)$, then $\mathbf{1}_K(z_1) = C(O(z_1)) = C(O(z_2)) = \mathbf{1}_K(z_2)$, so membership is fibre-constant. Conversely, (5) makes $C(y) = \mathbf{1}_K(z)$ for any z with $O(z) = y$ well defined and exact.

The fibre criterion concerns static classification — whether membership in a safe set factors through the observation — and applies verbatim to three distinct certification problems, which should not be conflated: constraint certification ($K = \mathcal{V}$), robust viability certification ($K = \text{RViab}(\mathcal{V})$), and policy-specific safety certification ($K = \text{Safe}_T^\Pi$). The same theorem governs all three, but its governance meaning differs. The criterion is the zero-step, label-valued analogue of the common-action test: an observation-based verdict assigns a single label to an entire fibre, exactly as an observation-based action assigns a single action to an entire belief, so both are selector problems. They remain distinct — a fibre may fail exact certification while a policy is still viable, and exact static certification does not imply dynamic viability; hence certification is treated as a layer separate from viability throughout.

Corollary 1 (safety-crossing fibres and the certainly-safe set). If two admissible states share an observation and lie on opposite sides of the *same* component safety constraint — with the safe set of that certification problem read as the single floor $K = \{q_j \geq 0\}$; or, in the general form of the hypothesis, one of the two states lies in K and the other outside K — no exact observation-only certificate exists. (Crossing one floor of an intersection $K = \bigcap_j \{q_j \geq 0\}$ does not by itself put the pair on opposite sides of the intersection — the state satisfying floor j may violate floor i — hence the single-floor scoping.) The largest set of observations that can soundly be labeled safe without further information is the certainly-safe set

$$\mathcal{Y}_{\text{safe}} = \{y \in O(Z) : O^{-1}(y) \subseteq K\}.$$

Proof sketch. A fibre containing one state in K and one outside violates (5), so Proposition 7 denies the certifier. On $\mathcal{Y}_{\text{safe}}$ every compatible state is safe, so the constant verdict “safe” is sound; outside it some compatible state is unsafe, so no sound “safe” verdict exists.

The governance reading: a composite index whose fibres cross a floor boundary cannot support an exact certification of that floor, however carefully it is thresholded; the certainly-safe set is the sound region where the index may certify safety, outside which it must fall silent. The complete discussion, with the fibre figure, is given in the Supplementary Material (S1).

4.2 The certainty-equivalence trap

The obstruction of Proposition 5 relies on a non-injective observation — information lost. The same emptying occurs with a fully *injective* observation for a single policy: the canonical certainty-equivalence controller that applies a perfect-information law to the uncorrected observation.

Remark 5 (the certainty-equivalence trap). Consider $\dot{S} = u - g(S)$ with $u \in [0, \bar{u}]$, g strictly increasing on $[S_{\min}, S^* + b]$, $g(0) = 0$, and $\mathcal{V} = [S_{\min}, S^*]$. Assume the range condition $g(S + b) \leq \bar{u}$ for every $S \in [S_{\min}, S^*]$, so that the controller below is admissible. Under perfect information the zeroing feedback $u(t) = g(S(t))$ gives $\dot{S} = 0$, so every $S_0 \in \mathcal{V}$ is viable and $\text{Viab}(\mathcal{V}; U, \pi_{\text{perf}}) = \mathcal{V} \neq \emptyset$. Now take the injective, biased observation $\hat{S} = S + b$ with $b > 0$, and the canonical certainty-equivalence controller: the perfect-information zeroing law applied directly to the uncorrected observation, $u = g(\hat{S}) = g(S + b)$. Then

$$\dot{S} = g(S + b) - g(S) > 0 \quad \forall S,$$

and since g is strictly increasing on the compact interval $[S_{\min}, S^* + b]$, $\dot{S}$ is bounded below by a positive constant there; hence S strictly increases and exits above S^* in finite time from every $S_0 \in \mathcal{V}$: the kernel of this single controller is empty. The bias-corrected controller $u = g(\hat{S} - b) = g(S)$ uses the observation map’s structure, restores $\dot{S} = 0$, and keeps the kernel nonempty.

The mechanism is a *failure of one policy to use the observation map’s structure*, not a loss of information: for a known invertible observation the output-feedback class is as powerful as the state-feedback class. No signalling or nonclassical-information analogy is needed for this known-bias policy example. The remark is policy-specific; it does not assert that the entire certainty-equivalence class empties the kernel.

5. The Sufficiency Landscape

For contrast, we record the sufficiency results against which the obstruction calculus is defined; the proofs are in the cited sources.

(a) Veliov’s output-feedback condition. Veliov (1993) gives a sufficient condition for the existence of an output-feedback regulation map under incomplete and inexact measurement, reducing under perfect measurement to the classical viability condition (Haddad, 1981). Veliov’s theorem certifies existence; the obstruction calculus certifies nonexistence; a problem satisfying neither requires a stronger observer or a finer observation structure.

(b) The estimation-tube programme. Cardaliaguet, Quincampoix, and Saint-Pierre (2007) survey the viability approach to differential games — viability and discriminating kernels for games with hard constraints, with value characterizations in Dini-derivative form. Under the set-membership semantics (Kurzhanski and Vályi, 1997), imperfect measurement is carried to an estimation space of sets of compatible states whose solution propagates the belief under a single control signal, so its controls are common controls; where that reduction applies with matching policy and admissibility hypotheses it is exact for values, and those hypotheses are not automatic. The common-action obstruction (Theorem 2) is the certificate that no jointly admissible selection exists at the information state in question (Section 6.3).

(c) Observer-and-buffer transfer. If a full-information feedback exists with a strict inward margin and an observer supplies exponentially convergent estimates, output feedback of the estimate can preserve safety on a buffered subset — the classical observer-to-control separation idea (Sontag, 1998) — provided the buffered set has been proved invariant under the

resulting input perturbations, the feedback’s Lipschitz constant and the observer’s error bound jointly covering the perturbation. No universal erosion coefficient or no-loss theorem holds without such a per-plant premise; the precise condition is in Supplementary Material, S2. Under margins and convergence, output feedback can work; Sections 3–4 state when it cannot.

(d) The linear substitution alternative. For a finite linear model in which a resource-typed system must meet a demand vector through specified substitution pathways, exactly one of the following holds (Farkas, 1902; Gale, 1960): (i) there exists a non-negative pathway vector $a \geq 0$ satisfying the linear substitution constraints; or (ii) there exist multipliers $\alpha, \beta, \gamma \geq 0$ with

$$\begin{aligned}\alpha^\top R + \beta^\top E - \gamma^\top Q &\geq 0 \quad \text{componentwise,} \\ \gamma^\top s^{\text{req}} &> \alpha^\top x + \beta^\top e.\end{aligned}$$

The second statement is a separation certificate that the pathways cannot meet demand within the typed resource and capacity bounds — the feasibility-side complement of the observation-side obstructions of Sections 3–4.

6. Discussion

6.1 The shape of the calculus

The entries below distinguish dynamic, epistemic, classification, and policy-specific mechanisms; they are not all information-theoretic nonviability certificates. Table 2 collects the certificates, the response correspondence each one empties, and the design consequence each one licenses.

The exit certificate (Theorem 5) is the dynamic obstruction: it defeats all controls, so it needs no observation argument. The epistemic-emptiness construction (Proposition 5) exhibits the mechanism at minimal size — one scalar state, two admissible controls, one constant observation — in the admissibility form of Section 3.6. The common-action obstruction (Theorem 2) is the static obstruction: it defeats all policies at a single information state. The delayed-information obstruction (Theorem 3) interpolates between them in time. The fibre criterion (Proposition 7) is the certification obstruction: it limits not policies but verdicts. The certainty-equivalence trap (Remark 5) limits a single uncorrected policy. Finite-horizon completeness (Theorem 1) supplies the exact finite-setting counterpart of these sufficient certificates.

A governance failure of observation can be diagnosed by which certificate applies. If the exit certificate holds, no monitoring design helps — the dynamics are adverse under every control. If only the common-action obstruction holds, the remedy is a finer observation structure (an informative measurement distinguishing the incompatible states). If the delayed-information obstruction holds with a slack timing bound, the remedy is a shorter review interval or a faster indicator. If the fibre criterion fails, the remedy is an observation that separates the sides of the floor rather than a better threshold on the aggregate.

6.2 Relation to barrier certificates

Barrier certificates certify safety from a scalar function whose zero level set separates the unsafe region from all trajectories (Prajna and Jadbabaie, 2004; Prajna, Jadbabaie, and Pappas, 2007); the converse, under convex-duality conditions on density functions, is Prajna and Rantzer (2005), and necessary-and-sufficient hybrid characterizations are Maghenem and Sanfelice (2019). Theorem 5 is the unsafety complement of that converse, not its observation-theoretic counterpart: a Dini-drift certificate of unsafety needing no observation structure, constructive in the enforcing disturbance and exit time. The remaining certificates (Proposition 5, Theorem 2, Theorem 3,

Proposition 7) act at the level of the information set itself: barrier constructions presuppose a state — or a specified state estimate — for feedback, whereas the common-action and timing obstructions certify the emptiness of the common safe-action set at a given information state, which barrier methods do not address.

6.3 Relation to the estimation-tube programme

The estimation-tube programme absorbs imperfect measurement into set-valued dynamics on an estimation space (Kurzbaniski and Vályi, 1997; Cardaliaguet, Quincampoix, and Saint-Pierre, 2007); the present theorems delimit its reach rather than contradict it. Where the reduction applies with matching policy and admissibility hypotheses it is exact for values, and its belief-state controls are common controls; the obstruction calculus is the local certificate layer of that programme — the common-action obstruction certifies that the belief-state predecessor is empty, the timing bound quantifies when it fails for reasons of information arrival, and the fibre criterion states when no observation-based verdict can be exact. Belief-space viability answers *whether*; the obstruction calculus answers *why not*, and *what to change*. A single worked system carrying the three methods side by side — barrier certificate, estimation-tube reduction, and obstruction certificate — is given in the Supplementary Material (S2).

6.4 Consequences for monitoring and indicator design

Five consequences follow directly from the certificates. Each translates a formal impossibility into a concrete design specification for monitoring systems under partial ecological information.

Timing. Theorem 3 gives a conditional sufficient bound on monitoring timing in the continuous model, not a general minimum frequency: the first informative observation must precede the enforced exit time $\inf q/\varepsilon$. A review interval longer than the worst-case time from the most-constrained compatible state — the minimum- q state, which has the least constraint slack — to constraint violation is an information structure under which the violation is undetectable in time, whatever the review then concludes. The bound is computable from the drift certificate (2), which a robustness analysis computes in any case.

Coarseness. Theorem 2 states that an observation structure merging two compatible states with incompatible safe controls is nonviable *at that information state*, even though both states are viable in isolation. The design consequence is that the observation must separate safety classes, not states: an indicator is exposed to Theorem 2’s obstruction when some observation fibre crosses the safe-control partition of the state space — merging states whose safe controls differ is what the theorem certifies, and keeping every fibre within a single class is what removes that exposure; refinement is never harmful (a finer observation merges nothing the coarser one did not), though Theorem 2 claims no benefit beyond the removal, and other mechanisms may still certify nonviability on a finer partition.

Sensor regularity. The manuscript does not posit one universal numerical sensor h, ρ . A proposed sensor regularity condition (R3) is model-specific: thresholded readings and tolerance-band readings can both fail closed-posterior/outer-limit or approximation obligations under their respective exact conventions. For any claimed regular-observation lemma, one must verify uniform joint approximation of reachable states, admissible actions, readings and exact posteriors, rather than infer posterior regularity from continuity of the raw sensor alone. These caveats do not themselves establish a failure for every sensor model.

Certificate	Emptied response correspondence	Design consequence (Section 6.4)
finite-time exit (Theorem 5)	the response set of some compatible state, even under full information	none — dynamics, not information
state-dependent admissibility example (Proposition 5)	$U^B(B) = \{0\}$, but $\mathcal{R}_V^B(B) = \emptyset$	enlarge the common safe-command set
common action and tube (Theorem 2, Proposition 2)	$\mathcal{R}_V^B(B) = \emptyset$, $\mathcal{A}_{\text{tube}}(B, \Delta) = \emptyset$	a separating observation; a shorter review interval
delayed information (Theorem 3, Remark 2)	$\sigma^*(B_0) < T_{\text{obs}}$	an earlier informative observation
fibre certification (Proposition 7, Corollary 1)	$K \neq O^{-1}(O(K))$	a refined index; per-floor measurement
certainty equivalence (Remark 5)	kernel of the uncorrected controller $u = g(\hat{S})$ is empty	correct the known bias
finite-horizon recursion (Theorem 1)	$B_0 \notin \mathcal{W}_N$	exact characterization (Section 3.1)

Table 2: The obstruction certificates, the response correspondence each one empties, and the design consequence each one licenses. The exit and common-action/timing rows concern nonviability in their stated systems or classes; the fibre row concerns verdicts and the certainty-equivalence row one uncorrected controller. The last row is the exact finite-horizon characterization.

Aggregation. Proposition 7 and Corollary 1 apply verbatim to composite indices, which are observation maps from the system state to a scalar. An index whose fibres cross a floor boundary cannot support exact floor certification; the certainly-safe set is the domain in which an index-based verdict is sound. Where separately-binding floors matter, exact certification requires an observation that separates every admissible pair on opposite sides of a floor (per-floor measurement is one sufficient design), a formal complement to the thesis, argued informally since Martinez-Alier, Munda, and O’Neill (1998), that weakly comparable values cannot be fused into one metric without loss. The requirement is sharpest where the floor constrains the productive base rather than its output: an aggregate of services can read at full level while the base regenerating them is reduced beneath it, and only an observation reaching the base itself can detect the discrepancy.

Bias. Remark 5 shows that a known biased indicator causes one uncorrected certainty-equivalence policy to fail on a system with a viable corrected policy; it does not empty the kernel of all output-feedback policies. Monitoring therefore has two obligations — bounding the observation error and supplying the correction — and the second lies entirely at the governance structure’s discretion: the indicator is published, the correction is not applied, and the kernel empties although the information was sufficient.

Institutions. The hierarchy of Section 2.1 records that epistemic contraction is one of three causes of kernel shrinkage, the others being disturbances and institutional restriction (authority, enforcement, allocation). The epistemic-institutional kernel $\text{IRViab}_{\mathcal{J}}(\mathcal{V})$ (Section 2.1) combines them: an institution restricted in what it may observe *and* in what it may command inherits both contractions. The obstruction calculus characterizes the first; its certificates remain valid, and typically tighten, when the institution’s command set is further restricted, since restriction of the action correspondence can only empty kernels, never create them (one-sided monotonicity of viability under correspondence restriction).

Proposition 8 (monotonicity of the epistemic kernel). The robust epistemic kernel is monotone in its constituents: enlarging the action sets, reducing the disturbance sets, refining the information structure, or enlarging the policy class never shrinks the kernel. Writing $\text{ERViab}^{\cdot}$ for the kernel with the indicated constituent, $U_1 \subseteq U_2$, $D_1 \subseteq D_2$, $\Pi_1 \subseteq \Pi_2$, and “ $\mathcal{I}_1$ refines $\mathcal{I}_2$ ” imply respectively

$$\begin{aligned} \text{ERViab}^{U_1} &\subseteq \text{ERViab}^{U_2}, & \text{ERViab}^{D_2} &\subseteq \text{ERViab}^{D_1}, \\ \text{ERViab}^{\Pi_1} &\subseteq \text{ERViab}^{\Pi_2}, & \text{ERViab}^{\mathcal{I}_2} &\subseteq \text{ERViab}^{\mathcal{I}_1}. \end{aligned}$$

This refinement inclusion compares kernels on the *same domain of prior beliefs*; it is not a literal inclusion of differently typed already-observed fibres.

Proof sketch. A policy viable under the smaller action sets, larger disturbance sets, smaller policy class, or coarser information structure remains admissible and nonanticipative under the corresponding enlargement or refinement, while the adversary’s options are unchanged or reduced; the same policy witnesses viability.

6.5 Limitations

The certificates are sound but not complete; several extensions are noted as open.

- (i) The continuous-time certificates of Sections 3.2–3.3 and 3.5–3.7 are sufficient conditions for nonviability, not necessary-and-sufficient characterizations of the epistemic kernel (backward recursion is complete in the finite-horizon finite-system setting, Section 3.1); the middle ground (neither a Veliov-type condition nor an obstruction certificate) is open. In continuous settings completeness would additionally require a measurable-selection theory beyond the explicit selections the certificates exhibit; completeness is therefore claimed only where backward recursion is exact (Section 3.1).
- (ii) The timing obstruction of Theorem 3 (and its threshold form, Remark 2) certifies failure by exit *before* the first informative observation. It is blind to the complementary mode of nonviability — *insufficient post-observation recourse* — in which a common blind-window control keeps every compatible branch safe, yet no observation-adapted control can recover every branch once the distinguishing observation arrives; the finite-horizon recursion (Section 3.1) captures that mode exactly, and the oracle-recourse certificate of Section 3.7 certifies it in the stated affine-state, affine-input and affine-facet class with exogenous disturbances — soundly against every measurable information-adapted policy of that plant class under an oracle relaxation of the post-observation class, with a worked three-branch instance whose delay threshold is exact; what remains open is a complete continuous-time characterization that does not relax the post-observation class. A discrete instance in which every one-step certificate is silent is given in the Supplementary Material (S3, A.3).
- (iii) The information-set semantics is set-membership; probabilistic or belief-space formulations require separate state-

ments, although the common-action obstruction transfers *mutatis mutandis* to any semantics in which the policy must choose a single action per information state.

- (iv) The base examples are finite-dimensional; the recourse certificate separately allows bounded measurable time-dependent coefficients for a fixed exogenous scenario. The continuous timing bound of Theorem 3 is sufficient under its strengthened adverse-realization hypothesis, not generally sharp; and delay systems require the retarded extension of the Dini comparison argument (Hale and Verduyn Lunel, 1993).
- (v) Theorem 5 is sound under its compatible original-system adverse-path hypothesis (H5.2). Neither a closed graph for D nor lower semicontinuity of ambient D alone proves that hypothesis; a convexified-inclusion trajectory is not an original-system witness. When the path premise is unverified, the theorem does not certify the model in question.
- (vi) The institutional consequences of Section 6.4 are design conclusions from the certificates, not empirical findings; their empirical testing belongs to applied studies.

7. Complete Characterizations and the Residual Gap

The certificates of Section 3 are sound sufficient conditions for nonviability, and Section 6.5 recorded that they do not exhaust the complement of the epistemic kernel. This section makes that gap precise: it exhibits two settings in which the viability question admits an exact answer — the one-step (finite discrete) setting, where the common-action certificate is necessary and sufficient, and the static-observation setting, where the problem reduces exactly to open-loop robust viability — and states the residual dynamic gap as an open problem.

Theorem 6 (one-step completeness). Let X, A, D, Y be finite, with transition $x^+ = F(x, a, d)$, observation $y = O(x, a, x^+)$, and safe set $\mathcal{V} \subseteq X$. Define the one-step safe set

$$S_1(x) = \{a \in U(x) : F(x, a, d) \in \mathcal{V} \text{ for all } d \in D(x)\}.$$

A post-observation cell $B \subseteq \mathcal{V}$ is one-step viable under $\mathcal{I}$ — some observation-based policy achieves $x_1 \in \mathcal{V}$ for every $x_0 \in B$ and every admissible disturbance — if and only if $\bigcap_{x \in B} S_1(x) \neq \emptyset$. Hence $B \notin \mathcal{W}_1$ exactly when the common safe-action set is empty, a separate finite/discrete analogue of the continuous Theorem 2.

Proof. ($\Rightarrow$) An observation-based policy depends on the record only, and every $x \in B$ produced the same record, so the policy plays a single $a \in U^B(B)$; safety of the first step forces $F(x, a, d) \in \mathcal{V}$ for all $x \in B$ and $d \in D(x)$, i.e. $a \in \bigcap_{x \in B} S_1(x)$. ($\Leftarrow$) Playing any a in the intersection keeps every successor in $\mathcal{V}$. $\square$

Combined with the backward recursion $\mathcal{W}_{k+1} = \text{Pre}(\mathcal{W}_k)$ of Theorem 1, Theorem 6 shows that the pair (*common-action certificate, recursion*) is a complete characterization at every finite horizon in finite systems: the certificate detects first-step emptiness, and the recursion propagates the obstruction to every later step.

Theorem 7 (static-observation reduction to open-loop viability). Let B_0 be a post-observation cell with one initial reading. Let $\mathcal{V}$ be compact with C^1 constraint functions q_j, f continuous with D compact-valued and of closed graph, $U(x) \equiv U$, and suppose the observation is *static*: $y(t) = O(x(t))$ is constant along every admissible trajectory, so no information accrues over time. Then every observation-based policy coincides with an *open-loop* control — a measurable function of time only, $u(\cdot) : [0, \infty) \rightarrow U$ — and $B_0 \in \text{ERViab}_{\mathcal{I}}(\mathcal{V})$ if and only if there

exists an open-loop control under which every trajectory from every $x_0 \in B_0$ and every admissible disturbance realization stays in $\mathcal{V}$ for all time. Within the constant-control subclass $u(\cdot) \equiv u_0$, the condition is the robust Nagumo boundary condition

$$\nabla q_j(x) \cdot f(x, u_0, d) \geq 0 \quad \text{for all } x \in \partial \mathcal{V}, \text{ active } j, d \in D(x),$$

a sufficient test (under the tangent-cone and solution-existence regularity of the robust Nagumo theorem) for invariance of $\mathcal{V}$ from *every* initial state. Under an appropriate constraint qualification the condition is also necessary for invariance of $\mathcal{V}$ as a whole, but it is *not necessary* for viability from a specified proper subset B_0 . A time-varying open-loop control may be viable when no constant control is; the open-loop reduction, not a global boundary test, characterizes viability from B_0 in the static class.

Proof. A static observation makes the record constant; for a fixed initial prior belief a record-based policy can therefore be represented by a time-only control on each initial observation cell. For a post-observation belief B_0 with one reading, this is one open-loop control, and the equivalence follows in both directions. Under the standard tangent-cone and existence hypotheses, the displayed boundary test suffices for invariance of all $\mathcal{V}$ with a constant control. It says nothing necessary about a smaller B_0 : for example $\mathcal{V} = [0, 2]$, $\dot{x} = x - 1$, $B_0 = \{1\}$ has a stationary viable trajectory although the velocity is outward at both ends of $\mathcal{V}$. $\square$

Theorem 7 makes the sense of completeness in the static class precise: the obstruction is exactly the nonexistence of a robustly viable open-loop control, and within the constant-control subclass the global boundary condition is a sufficient test for invariance of all $\mathcal{V}$, not a necessary test for viability from a specified B_0 . A finite obstruction-witness theorem is not supplied for the general static continuous class; when information accrues, post-observation recourse and timing add distinct modes.

The static-observation theorem localizes where completeness can fail, and the failure admits a precise structure. Consider a *two-phase model*: a window of K blind steps with declared blind class Π_B , followed by a specified reveal. Full physical-state revelation permits a matched full-information state kernel; revealing only a branch label may leave a nontrivial belief and requires the matching post-reveal belief kernel.

Proposition 9 (two-phase decomposition under a typed reveal). Fix a declared blind-window policy class and an exact post-reveal information model. A post-observation cell B_0 is viable for the two-phase game if and only if *one and the same* blind control keeps every compatible branch in $\mathcal{V}$ through step K and places every possible terminal information state in the kernel for the stated post-reveal policy class. Under full physical-state revelation this is the corresponding full-information *state* kernel; if only a branch label is learned and within-branch uncertainty remains, it is the appropriate post-reveal *belief* kernel. Neither case follows from a window-only failure test without these continuation hypotheses.

Proof. Restrict a viable policy to its blind part and then to each possible revealed information state: the same blind part must be safe and every continuation must be viable. Conversely, join a blind witness to the appropriate viable continuation after each distinguishable terminal record, provided the declared strategy class permits this nonanticipative pasting and the posteriors use the same admissible disturbance histories. This equivalence is a two-phase policy decomposition, not a finite dual certificate for arbitrary continuous observations.

Proposition 10 (no window-measurable pair is complete). Call a certificate pair *window-measurable* when its two verdicts depend only on the window sub-model: the initial belief, the branch transitions within the window, the floors, and the observation schedule. Then no window-measurable pair is complete for the two-phase model: the four-state (three safe-state) instance of Supplementary S3/A.3 and its variant agree on all window data, the certificates are silent in both, and yet one is nonviable (x_4 exits under every post-reveal action) while the other is viable (x_4 is maintained).

Proof sketch. The two instances differ only in x_4 's post-reveal transitions, so any window-measurable pair returns identical verdicts — here, silence on both. The exact recursion decides differently: in the first instance every post-reveal action exits $\mathcal{V}$, so $x_4 \notin \text{RViab}(\mathcal{V})$ and the belief misses $\mathcal{W}_2$; in the second x_4 is maintained, lies in the kernel, and the belief is viable. The two finite transition tables can be checked by exhaustive recursion (Supplementary S1, verification record). The residual dynamic gap of Open Problem 1 is thereby narrowed to a definite target: any complete certificate pair for the dynamic class must read data beyond the window sub-model.

Open Problem 1 (dynamic-observation completeness). Give a finite, checkable certificate that is necessary and sufficient for membership in $\text{ERViab}_{\mathcal{I}}(\mathcal{V})$ under dynamic observation beyond a suitably defined belief-space restatement below, or prove a separation showing that none exists in a natural class. Abstract transfinite fixed-point recursions do not alone provide a finite, checkable family of original-system obstruction certificates. With exact original-system reachability, disturbance rectangularity, compatible extension/gluing and implementable measurable belief policies, a belief-space viability kernel gives the matching policy statement; a generic set-valued reachability recursion need not equal the original policy kernel. Compact reachability and closed observation cells must be supplied if that kernel is formulated on compact beliefs. These conditions are not asserted for every measurable observation map in this paper.

8. The Selector Principle, Dual Certificates, and Monitoring Adequacy

The certificates of Section 3 are stated mechanism by mechanism. This section records the one object they consolidate, the exactness class in which its emptiness is also complete, and the design converse; it adds no new certificate and modifies none.

Typed selectors. For the finite-system recursion of Theorem 1, with $B \subseteq \mathcal{V}$, write

$$\Gamma_N^{\text{fin}}(B) = \{a \in U^B(B) : \text{Post}(B, a, y) \in \mathcal{W}_{N-1} \text{ for every possible } y\}.$$

This is a discrete predecessor. The continuous one-review held response is $\Gamma_{\Delta}^{\text{hold}}(B) = \mathcal{A}_{\text{tube}}(B, \Delta)$ with trajectory-wise safety and admissibility as defined above. An infinite-horizon held predecessor additionally asks that every exact post-review belief lie in a specified held-policy viability set. It is not Theorem 1's finite-system recursion until a sampled-transition embedding, information and disturbance rectangularity, and policy implementability are supplied.

Scope of witnesses. An original-system boundary-exit path (under its path hypotheses), an admissible Farkas infeasibility witness, or a negative exogenous recourse scenario proves only the nonviability assertion in its declared model and policy class. A crossing observation fibre proves failure of an *exact static certifier*, not policy nonviability. The biased-reading example tests one uncorrected policy, not all output-feedback policies. These mechanisms must not be combined into one untyped dual pair.

Exact and open cases. The finite-state finite-horizon predecessor is exact under its stated observation/strategy semantics. A particular polytope LP has its own feasibility–Farkas alternative, not a global continuous epistemic-kernel duality. Static observation permits an open-loop policy reduction in Theorem 7; it does not by itself produce a finite obstruction witness for every nonviable belief. The grid of Remark 6 classifies its declared whole-window hold instance only. For a general continuous recursive class, the converse “every nonviable belief has a witness from this finite checkable certificate family” remains Open Problem 1. An abstract transfinite predecessor identity is not a finite checkable certificate.

Monitoring adequacy (the design converse). Obstruction answers why nonviability is forced; design asks what must be measured. In the one-step exact class the converse is immediate: an observation structure is adequate on a target exactly when every fibre meeting the target has a nonempty common safe action — a discrete common-action test distinct from the static-verdict fibre criterion of Proposition 7; pairwise intersection is necessary and, off convexity, not sufficient (the three two-point codices of the companion monitoring paper). Four rules follow as identities on the audited systems. (i) *Timing*: on the delayed hidden-regime *whole-window hold* class, viability at integer deadline T_{obs} holds exactly when $T_{\text{obs}} \leq \sigma_{\text{hold}}^*(B_0) = \lfloor z_0 - 1 \rfloor$, equivalently $z_0 \geq 1 + T_{\text{obs}}$; the boundary case $z_0 = 1 + T_{\text{obs}}$ is viable — the review must land no later than worst-case exit (Section 9's grid; the strict form fails at the boundary cell, as the verification script records). (ii) *Coarseness*: a fibre merging states with disjoint safe-action sets obstructs at that information state while every branch remains viable (Theorem 2); the remedy is a fibre split, not a finer threshold on the same index. (iii) *Aggregation*: an index whose fibres cross a floor cannot support an exact safe/unsafe verdict; report the certainly-safe set or fall silent (Corollary 1). (iv) *Bias*: the trap of Remark 5 empties a nonempty kernel under the uncorrected law; the remedy is structural correction of the observation map, not a better sensor. The consolidated statement, with every audited count cited to its system, is maintained in the supplementary material.

9. A Symbolic Case Study and a Coverage Audit

We instantiate the three observation-side mechanisms on a concrete resource model and audit the certificates against the exact recursion of Theorem 1. The model is a single-species stock $S \in [0, 1]$ (fraction of carrying capacity) with discrete dynamics

$$S^+ = \text{clip}(S + S(1 - S) - u + w), \\ u \in [0, 0.3], \quad w \in \{-0.02, 0, +0.02\},$$

harvest u , recruitment noise w read adversarially, and a sustainability floor $S_{\min} = 0.3$ (Clark, 1990).

The fibre criterion. The regulator observes a coarse two-bin index, low if $S < 0.6$ and high if $S \geq 0.6$. The safe set $\mathcal{V} = [0.3, 1]$ is not a union of index bins: the low bin contains both 0.1 (below the floor) and 0.4 (above it). By Proposition 7 no exact observation-only certifier exists, and by Corollary 1 the certainly-safe set is the single bin $\mathcal{V}_{\text{safe}} = \{\text{high}\}$. A verdict of “safe” on the strength of a low reading is therefore unsound; the index can certify safety only when it reads high.

Two-dimensional aggregation. The fibre criterion also governs a multidimensional stock. Take a two-species state $(S_1, S_2) \in [0, 1]^2$ with the floor $S_1 \geq 0.4$, and suppose the regulator observes only the aggregate $I = S_1 + S_2$. The states $(0.5, 0.5)$ and $(0.2, 0.8)$ are both compatible with the reading $I = 1.0$, yet the first satisfies the floor and the second violates it, so by Proposition 7 no exact observation-only certifier exists;

by Corollary 1 the certainly-safe readings are exactly $I \geq 1.4$ (the smallest S_1 compatible with I is $I - 1$, so $I - 1 \geq 0.4$). An aggregate reading below 1.4 must therefore fall silent, however carefully thresholded — the two-dimensional form of the aggregation consequence of Section 6.4.

The delayed-information timing bound (declared hold class). Regimes $\theta \in \{-1, +1\}$ follow $z^+ = z + \theta u$, $u \in \{-1, +1\}$, with floor $z \geq 1$. Here the institutional class fixes *one action for the entire* T_{obs} -step blind window; it may select a regime-specific action only after the reveal. At $z_0 = 2$, either hold drives one branch down at one unit per step, giving one safe blind step; after reveal $u = \theta$ restores growth. Thus the hold-class timing certificate fires for integer $T_{\text{obs}} > 1$. This is an exact *discrete hold-class* calculation from the finite recursion, not a verification of the continuous Theorem 3: $+1, -1, +1, \dots$ can keep both branches safe from $z_0 \geq 2$ without an informative observation. In the fishery above the same certificate is silent in the normal regime because $u_0 = 0$ keeps every stock above the floor.

The certainty-equivalence trap. With a biased index $\hat{S} = S + 0.1$, $g(S) = S^2$, and $\mathcal{V} = [1, 2]$, the certainty-equivalence policy $u = g(\hat{S})$ produces $\dot{S} = g(\hat{S} + 0.1) - g(\hat{S}) = 0.2\hat{S} + 0.01 > 0$, which reaches the upper bound from $S = 1$ in ≈ 3.3 time units, while the bias-corrected policy $u = g(\hat{S} - 0.1)$ keeps $\dot{S} = 0$ and the kernel nonempty (Remark 5).

A reproducible delayed hidden-regime model, with its policy class. The state is (z, θ) , $\theta \in \{-1, +1\}$, $z \geq 0$; $z^+ = z + \theta u$, $u \in \{-1, +1\}$, and $\mathcal{V} = \{z \geq 1\}$. No regime is observed before the integer horizon T_{obs} ; it is revealed exactly then. The *audited institutional class* is $\Pi_{\text{hold}} = \{u_0 = \dots = u_{T_{\text{obs}}-1} = +1 \text{ or } -1\}$, with regime-adapted actions after the reveal. This is a whole-window hold, *not* all per-step sequences over $\{-1, +1\}$. From a known z_0 , the two held histories after $k < T_{\text{obs}}$ steps are $z_0 - k$ and $z_0 + k$. The maximal integer number of jointly safe blind steps is $\sigma_{\text{hold}}^*(B_0) = \lfloor z_0 - 1 \rfloor$; its linearly interpolated crossing time is $z_0 - 1$. Once revealed, $u = \theta$ keeps the surviving branch safe forever. Consequently viability *in this hold class* is exactly $z_0 \geq 1 + T_{\text{obs}}$. The one-step certificate covers $z_0 < 2$; the hold-class timing bound covers $z_0 \geq 2$ and integer $T_{\text{obs}} > z_0 - 1$. Under the broader per-step class, $+1, -1, +1, \dots$ keeps both branches safe from any $z_0 \geq 2$, so the hold-class timing verdict does not transfer to that class.

A coverage audit. To quantify the gap identified in Section 6.5, we audit the certificates against the exact recursion *restricted to the declared whole-window hold class* on the delayed hidden-regime system, over the grid $z_0 \in \{1.0, 1.1, \dots, 2.5\}$, $T_{\text{obs}} \in \{1, 2, 3\}$. Table 3 and Figure 3 report the true verdict, the one-step certificate, and the timing bound. The archived `paper2_coverage_audit.py` reproduces the finite-grid classification. This discrete calculation is not a proof of the continuous Theorem 3.

Remark 6 (joint completeness in the delayed class). In the *declared whole-window hold class*, of the 42 nonviable grid cells, 30 are classified by the finite one-step test and the remaining 12 — those with $z_0 \geq 2$ and integer $T_{\text{obs}} > z_0 - 1$ — by the discrete hold-class survival calculation; none is uncovered. This is joint completeness *only in that restricted class*. For comparison, in the broader per-step class only the 30 common-action cells are nonviable; the 12 hold-only timing cells are viable by alternating controls.

Why joint completeness holds in this hold class. The hold-class viability condition is $z_0 \geq 1 + T_{\text{obs}}$ at integer horizon. The finite one-step test covers $z_0 < 2$; the exact discrete hold-class calculation covers $2 \leq z_0 < 1 + T_{\text{obs}}$, since a whole-window hold then sends one branch below the floor. These disjoint sets partition exactly the hold-class nonviable cells. This partition is *not* a completeness result for the unrestricted policy class, where switching within the blind window changes the verdict.

Calibration. The certificate parameters correspond to quantities estimable from standard stock-assessment practice: the drift margin ε is a net growth rate at the floor; the first informative observation time T_{obs} is the assessment or review periodicity; the bias b of the certainty-equivalence instance is the systematic error of the indicator; and the floor is the reference point of the management rule. Calibrating these four numbers from a published assessment is all the machinery the certificates require; the present section stays symbolic, and the empirical calibration is left to applied studies.

A two-patch protection audit. At each step one of two patches $z_i \in \{0, 1, 2, 3\}$ receives protection $u \in \{1, 2\}$. The transition is $z_i^+ = \min\{3, z_i + 1 - 21[u \neq i]\}$: the protected patch gains one unit up to the cap, while the other loses one net unit (recruitment of one less a loss of two). The floor is $z_i \geq 1$ and the observed reading is $y = z_1 + z_2$. In this finite discrete model the exact one-step test is Theorem 6, not the continuous-time boundary tangency condition of Theorem 2. The actual current y-fibres have verdicts in Table 4. The $y=4$ fibre is especially instructive: all three states are safe and individually viable under full information, so $y=4$ is a certainly-safe reading, yet the fibre is not viable. Action 1 fails at $(3, 1)$, sending it to $(3, 0)$; action 2 fails at $(1, 3)$, sending it to $(0, 3)$. The $y=3$ fibre has the analogous action conflict. The $y=2$ singleton $(1, 1)$ is nonviable even with full information: it is a dynamic, not an epistemic, failure. The viable $y=5$ fibre has a history-dependent witness: play $u = 1$ while the reading remains 5, reaching the known singleton $(3, 1)$ when it first becomes 4; thereafter alternate 2, 1 to cycle $(3, 1)$ and $(2, 2)$. The $y=6$ singleton reaches $y=5$ under $u = 1$ and uses the same continuation. Certainly-safe readings are $y \geq 4$, whereas viable *full initial fibres* have $y \geq 5$. These exact recursion results concern the declared finite model, not a calibrated resource system.

The audit is symbolic, including two two-dimensional instances (the aggregated reading above and the two-patch protection audit); a full multidimensional belief-space numerical campaign is deferred to future work, and the case study makes no claim of a general-purpose computational calculus.

10. A Probabilistic Development: Belief-State Safety Values

The certificates of Sections 3–4 rest on set-membership semantics: the information state is a set of compatible states. This section records the stochastic counterpart — the belief-state safety value of partially observable control — and its obstruction certificate, connecting the calculus to the partially observable literature (Åström, 1965; Smallwood and Sondik, 1973).

Let X, A, D, Y be finite, with a stochastic transition $T(x' | x, a)$ and an observation likelihood $g(y | x, a, x')$. Augment the state space with an absorbing unsafe state $\perp$: for $x \in \mathcal{V}$, $T(\perp | x, a) = 1 - \sum_{x' \in \mathcal{V}} T(x' | x, a)$, and $T(\perp | \perp, a) = 1$. A belief is a distribution b over $X \cup \{\perp\}$, updated by Bayes' rule $\tau(b, a, y)$, with $\mathbb{P}(y | b, a) = \sum_{x, x'} b(x) T(x' | x, a) g(y | x, a, x')$. The *safety value* is the maximal survival probability

$$V_k(b) = \max_{\pi} \mathbb{P}_{\pi}(x_t \in \mathcal{V} \text{ for } t = 0, \dots, k | b).$$

Theorem 8 (belief-state safety value). $V_0(b) = b(\mathcal{V})$ (the horizon counts the current state), and for $k \geq 0$

$$V_{k+1}(b) = \max_{a \in A} \sum_{y \in Y} \mathbb{P}(y | b, a) V_k(\tau(b, a, y)),$$

with the time convention that the policy acts, observes y , then acts again. The recursion is exact on every finite horizon; it is the Smallwood–Sondik belief-state value iteration specialized to the absorbing safety reward (Smallwood and Sondik, 1973).

T_{obs}	initial position z_0															
	1.0	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2.0	2.1	2.2	2.3	2.4	2.5
1	•	•	•	•	•	•	•	•	•	•	○	○	○	○	○	○
2	•	•	•	•	•	•	•	•	•	•	*	*	*	*	*	*
3	•	•	•	•	•	•	•	•	•	•	*	*	*	*	*	*

Table 3: Coverage audit on the delayed hidden-regime system *restricted to one action held throughout the blind window*: ○ viable; • nonviable, certified by the finite one-step common-action test (Theorem 6); * nonviable in the hold class, classified by the discrete hold-class survival threshold and exact finite recursion (not by the continuous Theorem 3). All hold-class nonviable cells are covered. The * cells can be viable under per-step switching.

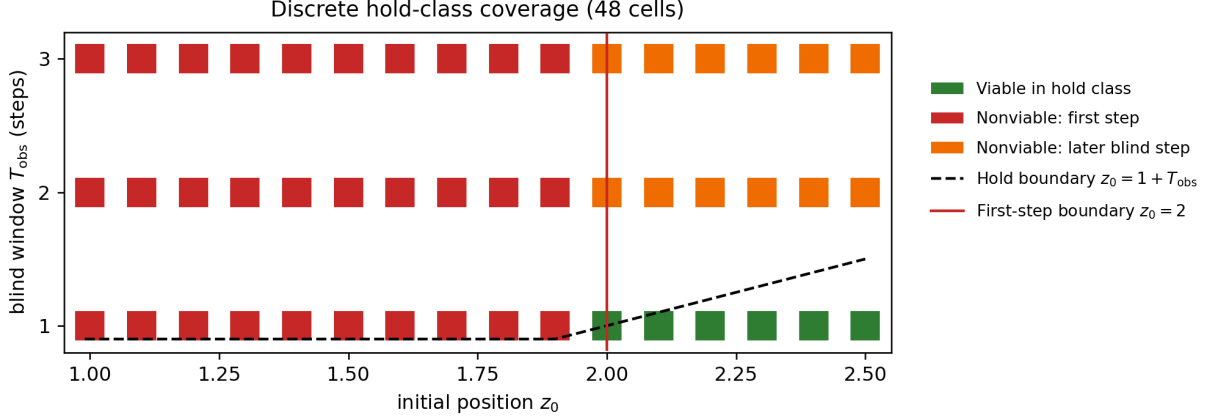


Figure 3: Whole-window hold-class coverage of Table 3: green = viable; red = nonviable under either class, diagnosed by the discrete one-step test; orange = nonviable *only in the audited hold class*, established by the discrete hold-class survival calculation. With per-step alternating actions, the orange cells are viable. The dashed curve is the *hold-class* boundary $z_0 = 1 + T_{\text{obs}}$, not an unrestricted boundary; the solid vertical line marks the one-step boundary $z_0 = 2$.

actual current fibre	y	verdict	certificate / witness
$\{(1, 1)\}$	2	nonviable	full-information one-step failure
$\{(1, 2), (2, 1)\}$	3	nonviable	no common safe first action
$\{(1, 3), (2, 2), (3, 1)\}$	4	nonviable	no common safe first action
$\{(2, 3), (3, 2)\}$	5	viable	$u = 1$ until $y=4$, then 2,1 cyclically
$\{(3, 3)\}$	6	viable	$u = 1$, then $y=5$ witness

Table 4: The five actual safe-state fibres of the aggregate reading $y = z_1 + z_2$. The one-step discrete obstruction is Theorem 6; the $y=2$ singleton fails even with full information. The $y=4$ reading certifies current safety but cannot support one common safe action. The viable rows admit the stated history-dependent policies.

Proof. Standard value iteration for the belief-state Markov decision process: conditioning on the observation after the action and applying the optimal continuation gives the identity, and the absorbing state makes V_k the k -step survival probability (Smallwood and Sondik, 1973). $\square$

Proposition 11 (chance-constrained common-action obstruction). Let $b \in \Delta(\mathcal{V})$ (so $b(\mathcal{V}) = 1$) and write $p(x, a) = 1 - \sum_{x' \in \mathcal{V}} T(x' | x, a)$. If

$$\sum_{x \in \mathcal{V}} b(x) p(x, a) \geq \delta \quad \text{for every } a \in A,$$

then $V_k(b) \leq 1 - \delta$ for every $k \geq 1$, and the chance constraint $\mathbb{P}_{b, \pi}(x_0, \dots, x_k \in \mathcal{V}) \geq 1 - \varepsilon$ is infeasible under every policy whenever $\varepsilon < \delta$.

Proof. The one-step survival probability under action a is $1 - \sum_x b(x) p(x, a) \leq 1 - \delta$, so $V_1(b) \leq 1 - \delta$. Since $V_{k+1} \leq V_k$ — more steps cannot raise the survival probability — $V_k(b) \leq V_1(b) \leq 1 - \delta$. $\square$

Proposition 12 (deterministic-kernel limit). Let X, A, Y be finite with $X = \mathcal{V} \cup \{\perp\}$, where $\perp$ is the unique absorbing unsafe state in *both* models. In the set-valued model of Section 3.1, assume every admissible disturbance gives the *same* successor $F(x, a, d) = F(x, a)$ (equivalently, take a singleton disturbance set), and its observation is independent of d : $O(x, a, d, x^+) = O(x, a, x^+)$. Use this deterministic single-successor transition and deterministic observation in the stochastic model as well. Suppose $T(x' | x, a) = \mathbf{1}[x' = F(x, a)]$, the observation likelihood is the corresponding point mass, and every action in A is admissible at each state under consideration, so the policy class in the stochastic value V_k agrees with the set-valued recursion $\mathcal{W}_k$. Let b have full support $B = \text{supp}(b) \subseteq \mathcal{V}$. Then for every possible observation the posterior support is exactly $\text{Post}(B, a, y)$. For the unrestricted sequential policy class, $V_k(b) = 1$ if and only if $B \in \mathcal{W}_k$, and if $B \notin \mathcal{W}_k$ then $V_k(b) \leq 1 - \min_{x \in B} b(x) < 1$. This quantitative bound does *not* follow merely because a stochastic transition has the same support as a set of possible adversarial disturbances; such a fixed stochastic kernel is governed instead by its transition probabilities and Proposition 11.

Proof. Under the point-mass transition and observation kernels, each initial state and deterministic observation-based policy generate a unique path. The posterior support is precisely the deterministic set-valued post-state. By the finite-horizon recursion of Theorem 1, $B \in \mathcal{W}_k$ if and only if some such policy keeps *every* state in B safe through step k , equivalently $V_k(b) = 1$ because all $b(x) > 0$ on B . If $B \notin \mathcal{W}_k$, every policy loses at least one initial state $x \in B$ by some stage $t \leq k$; its loss probability is therefore at least $b(x) \geq \min_{z \in B} b(z)$. Maximizing expected survival over policies gives the displayed bound. A multi-step failure need not occur on the first action. $\square$

Proposition 12 is the sense in which the chance-constrained obstruction degenerates to the common-action obstruction: at the deterministic limit $V_1(b) = 1$ exactly when the common safe-action set is nonempty, and the deficit from 1 is at least the smallest compatible mass $\min_{x \in B} b(x)$. The unrestricted deterministic-kernel bound is attained by the finite-action two-floor conflict at $V_k(b_0) = \frac{1}{2}$ for all k (Supplementary S1). The delayed hidden-regime grid of Section 9 is a *different, held-policy-class* calculation: its class-relative safety values also have half-mass deficits, but they are not values of the unrestricted V_k or direct instances of this proposition. Supplementary S1 reports its 48-cell table with that policy restriction explicit.

11. Conclusion

Under standard finite-dimensional regularity assumptions, viability under perfect measurement has a mature kernel, tangency, and approximation theory. Viability under incomplete observation has long been asymmetric: the sufficiency side is carried by Veliov’s output-feedback condition and the estimation-tube programme, while the failure side has received less systematic treatment. This paper supplies the missing instrument: the obstruction calculus develops sound, computationally interpretable certificates for nonviability — that is, sufficient conditions for nonviability, equivalently necessary conditions for viability. They do not exhaust the complement of the epistemic kernel (Section 6.5), and the paper claims no complete characterization. The mechanisms are organized by one selector principle (Proposition 1) — observation-based viability is the existence of a common recursively safe response — and are nested in an obstruction ladder (Proposition 3); in finite systems backward belief recursion is sound and complete (Theorem 1), so the calculus is exact where it can be and sound elsewhere.

The results must be separated by claim type. The exit certificate is dynamic and defeats every control; the common-action and conditional delayed-information certificates concern failure of a shared response; Proposition 5 is a nonconvex state-dependent *safety* instance with a nonempty common admissible set. The fibre criterion limits exact observation-only verdicts rather than viability, and the certainty-equivalence example defeats one uncorrected policy, not every observation-based policy. The finite recursion is an exact algorithm in its specified finite class. Each certificate that admits a design remedy identifies it: a faster indicator, a separating observation, an observation that separates the sides of each floor instead of aggregate certification, or the correction of a known bias. For sustainability governance, where observation is conducted through coarse indicators at discrete reviews, the calculus states the quantitative limits of what monitoring can deliver — and converts each impossibility into a design specification. The principal further elements are: finite one-step recursion is exact and static observation admits an open-loop policy reduction, without a general finite witness for nonviability; the dynamic finite-certificate gap remains open (Section 7); typed selector correspondences distinguish finite recursion, held continuous review and static verdicts, with conditional design rules (Section 8); a symbolic case study and a coverage audit reproduce the exact *whole-window hold-class* verdict on a delayed hidden-regime grid, with a two-patch audit carrying aggregation to two dimensions (Section 9); and the probabilistic belief-state lift connects the calculus to partially observable control, with unrestricted values distinguished from the separate hold-class values tabulated on that grid (Section 10 and Supplementary S1). A separate structural result concerns polytope-declared blind-window classes: the specified affine timing-feasibility problem is exactly computable as a linear program, while its relaxation of a different implementable hold class can have an unbounded survival gap (Section 3.3), and the residual dynamic gap de-

composes exactly, its post-observation recourse verdict not determined by any window-only certificate pair (Section 7).

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Declarations

Funding. None. **Competing interests.** None.

Data availability. No external data were used.

Code availability. The coverage-audit script `paper2_coverage_audit.py` is publicly available at <https://github.com/MIKEAA2020/general-sustainability> (folder `arena agent 1/paper rewrites/latex`); it regenerates Table 3 and Figure 3 verbatim. **AI declaration.** GLM (Z.ai), Qwen (Alibaba Cloud) and DeepSeek AI assisted with drafting and iterative review. The author reviewed and edited outputs and takes responsibility for the final work.

Author contributions. A.A. conceptualized the entire work, wrote, reviewed and edited the manuscript.